

Advancing Finger Vein Recognition: A Comparative Analysis of Feature Extraction and Matching

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ABSTRACT

Finger vein recognition has become a popular research topic in recent years. In this study, a new texture-based descriptor is introduced for systems that do not have much computing power to spare. The main idea is straightforward: instead of relying on a single neighborhood size, three of them—(8,1), (16,1), and (8,2)—are combined inside one sliding window. This design allows the descriptor to capture both fine and coarse texture details at the same time. The combined operator is denoted as $\text{LBP}_{(8,1)(16,1)(8,2)}^{\text{riu2}}$. After feature extraction, the behavior of the descriptor is examined in combination with different dimensionality-reduction methods, including Principal Component Analysis (PCA), Two-Dimensional PCA (2DPCA), and Two-Directional Two-Dimensional PCA ($2D^2\text{PCA}$). Experiments are carried out on three commonly used datasets: FV-USM, MMCBNU-6000, and UTFVP. For UTFVP, a careful preprocessing step with an accurately defined ROI is applied, since this dataset is more sensitive to registration and alignment. The system is tested by two different fusion strategies and three different protocols. Across all experiments, the fused-radius LBP and its reduced variants show strong performance and often surpass many traditional hand-crafted approaches. Additional comparisons with a modified AlexNet model and other state-of-the-art methods substantiate the effectiveness of the proposed descriptor. To show how the methods perform across different ranks, Cumulative Match Characteristic (CMC) curves are provided.

Keywords: Finger Vein Identification, Biometrics, Feature Extraction, Fusion, Dimensionality Reduction, Local Binary Patterns (LBP)

ÖZ

Parmak damar tanıma son yıllarda giderek daha popüler bir araştırma konusu hâline gelmiştir. Bu çalışmada, hesaplama gücü sınırlı olan sistemler için yeni bir doku tabanlı betimleyici önerilmektedir. Temel fikir oldukça basittir: tek bir komşuluk boyutuna bağlı kalmak yerine, üç farklı yapı—(8,1), (16,1) ve (8,2)—tek bir kaydırma penceresi içinde birleştirilmektedir. Bu tasarım, betimleyicinin hem ince hem de kaba doku ayrıntılarını eşzamanlı olarak yakalamasına imkân tanır. Birleştirilmiş işlemci $LBP_{(8,1)(16,1)(8,2)}^{riu2}$ şeklinde gösterilmektedir. Öznitelik çıkarımının ardından, betimleyicinin davranışı; Temel Bileşen Analizi (PCA), İki Boyutlu PCA (2DPCA) ve Çift Yönlü İki Boyutlu PCA ($2D^2PCA$) gibi farklı boyut indirgeme yöntemleriyle birlikte incelenmiştir. Deneyler, yaygın olarak kullanılan üç veri kümesi—FV-USM, MMCBNU-6000 ve UTFVP—üzerinde gerçekleştirilmiştir. UTFVP veri kümesi, hizalama ve kayıttan daha fazla etkilendiği için, bu veri kümesine dikkatli bir ön işlem adımı ve doğru tanımlanmış bir ROI uygulanmıştır. Sistem, iki farklı füzyon stratejisi ve üç farklı protokol altında test edilmiştir. Tüm deneyler boyunca, birleştirilmiş yarıçaplı LBP ve azaltılmış varyantları güçlü bir performans göstermiş ve çoğu geleneksel el yapımı temelli yöntemi geride bırakmıştır. Değiştirilmiş bir AlexNet modeli ve diğer güncel yöntemlerle yapılan ek karşılaştırmalar da önerilen betimleyicinin etkinliğini desteklemektedir. Yöntemlerin farklı sıralama seviyelerindeki performanslarını göstermek amacıyla Kümülatif Eşleşme Karakteristiği (CMC) eğrileri sunulmuştur.

Anahtar Kelimeler: Parmak Damar Tanıma, Biyometri, Öznitelik Çıkarımı, Füzyon, Boyut İndirgeme, Yerel İkili Örüntüler (LBP)

... *Dedicated to my parents*

For my parents, whose unwavering love and support made this
possible.

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LIST OF ABBREVIATIONS

$(2D)^2$ PCA	Two-Directional Two-Dimensional Principal Component Analysis
2DPCA	Two-Dimensional Principal Component Analysis
3D	3 Dimensions
3DFVSNet	3D Finger Vein Spherical Network
BGR	Blue-Green-Red
BN	Batch Normalization
CLAHE	Contrast Limited Adaptive Histogram Equalization
CMC	Cumulative Match Characteristic
CNN	Convolutional Neural Network
Conv Layer	Convolution Layer
CPBFL-BCL	Category-Preserving Binary Feature Learning with Binary Clustering
DBN	Deep Belief Network
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DoGHE	Difference of Gaussian-Histogram Equalization
EER	Equal Error Rate
Eq.	Equation
Eval Model	Evaluation Mode
FAR	False Accept Rate
FC	Fully Connected Layer
FC Layer	Fully Connected Layers
FedFV	Federated Learning Framework for Finger Vein authentication
FNIR	False Negative Identification Rate
FPIR	False Positive Identification Rate

FRR	False Reject Rate
FTA	Failure to Acquire
FTC	Failure to Capture
FTE	Failure to Enroll
FV-USM	Finger Vein Universiti Sains Malaysia Dataset
FVR	Finger Vein Recognition
GWO	Grey Wolf Optimization
HistEq	Histogram Equalization
HMI	Human-Machine Interface
HSV	Hue-Saturation-Value
L1	Manhattan Distance
LBP	Local Binary Pattern
LBP ^{riu2}	Rotation-Invariant Uniform Local Binary Pattern
LDP	Local Directional Pattern
LPQ	Local Phase Quantization
MMCBNU-6000	Multimodal Biometric Dataset from Chonbuk National University
NIR	Near-Infrared
NGA	Niching Genetic Algorithm
OneCycleLR	One Cycle Learning Rate Scheduler
PIN	Personal Identification Number
PCA	Principal Component Analysis
PDA	Personal Digital Assistant
PWBDC	Principal Wavelet-Based Directional Coding
px	Pixel
ReLU	Rectified Linear Unit
RGB	Red Green Blue

ROI	Region of Interest
SIFT	Scale-Invariant Feature Transform
SDBCCML	Steepest Deep Bipolar Cascade Correlative Machine Learning
SGD	Stochastic Gradient Descent
SURF	Speeded-Up Robust Features
SVM	Support Vector Machine
UTFVP	University of Twente Finger Vein Pattern Dataset
WLD	Weber Local Descriptor (texture feature)
Z-score	Standard score normalization method

Chapter 1

INTRODUCTION

Reliable identity checks matter in a lot of everyday systems, but the usual tools—passwords, PINs, and physical tokens—break down in very ordinary ways: people forget them, reuse them, share them, or lose them, and attackers can steal or copy them. Biometric recognition is meant to sidestep those weak points by tying authentication to measurable human characteristics, so access depends less on what someone knows or carries and more on what can be captured from the person at the moment of use. This thesis focuses on finger vein recognition, where identity is inferred from vascular patterns beneath the skin that appear in near-infrared images. The attraction of finger veins is practical as well as conceptual: the evidence is internal (so it is less affected by surface wear), and the overall system can be described in a straightforward pipeline—image acquisition, preprocessing and ROI localization, feature extraction, and matching. The remainder of this chapter introduces finger vein recognition from this system viewpoint and then narrows to the feature representations and evaluation protocols used throughout the thesis.

1.1 Finger Vein Recognition Systems

Finger vein recognition is a biometric approach that bases identity decisions on the internal vein pattern of the finger rather than on visible surface features. Interest in this technique has increased markedly over the last decade, largely because the vascular pattern is hidden beneath the skin and is therefore difficult to forge or steal [1].

To capture this information, the sensor illuminates the finger with near-infrared (NIR) light, typically in the wavelength range of 700–1000 nm. Hemoglobin in the blood absorbs much of the NIR radiation, while the surrounding tissue is comparatively translucent. In the recorded image, the veins therefore appear as dark lines against a brighter background, and the vascular network can be extracted with high contrast [2].

Finger vein identification also possesses several preferred attributes as a biometric modality, being universal, unique, permanent, and acceptable to users. In contrast to other biometric traits such as face, gait, or fingerprint, it possesses two very great advantages in particular [3]:

- **Liveness detection:** Because only veins of a living finger will absorb infrared light in the expected way, it is robust against spoofing with artificial replicas or pictures.
- **Internal structure:** The structure of veins is beneath the skin and therefore it is extremely difficult to fake or manipulate and more durable against damage or wear on the surface.

All these features render finger vein recognition especially suitable for high-security application, evoking increasing interest from academia as well as industry.

A typical finger vein recognition system comprises four major stages, as shown in Figure 1.1: image acquisition, preprocessing, feature extraction, and comparison. Early research laid the foundation for the field by demonstrating that vein patterns in the finger were distinctive and stable in large populations. Feature extraction techniques in early research included line-tracking algorithms to trace the direction of the vein pixel-by-pixel, and curvature analysis-based centerline extraction of various

vein widths and brightness.

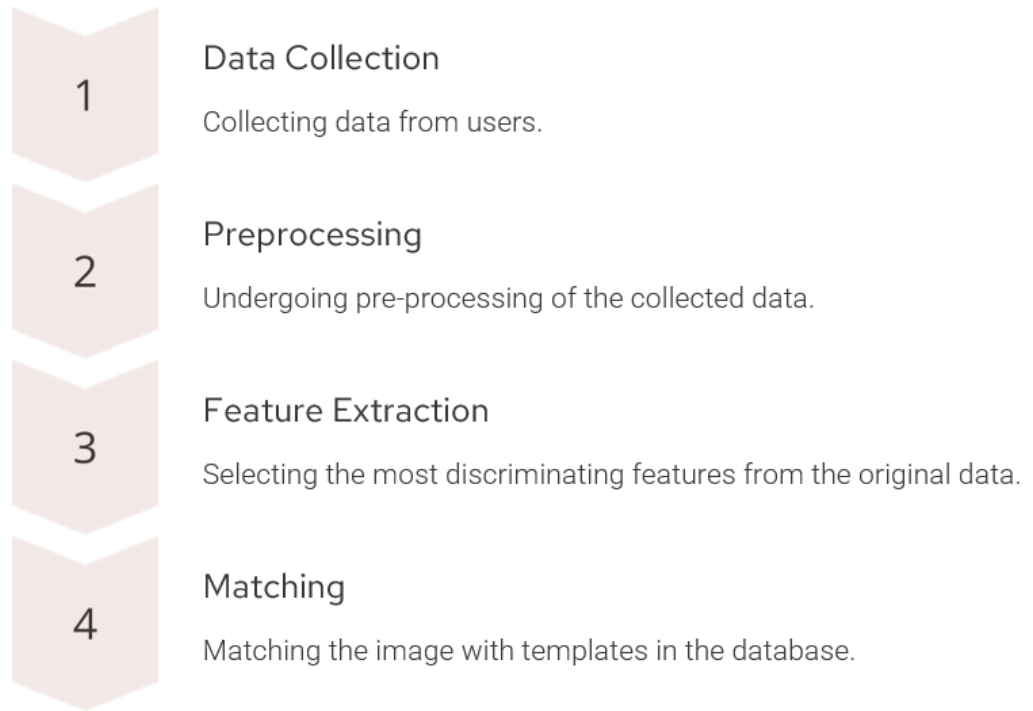


Figure 1.1: Standard Workflow for Finger Vein Recognition

The region has seen significant maturing over the past decade, from traditional hand-crafted methods (such as Gabor Filters) to deep learning methods. While CNNs (e.g., AlexNet, DenseNet, ResNet) now dominate state-of-the-art systems with implicit learning of discriminative vein patterns, work today extensively investigates their relative merits over traditional methods. Studies compared CNN-based methods rigorously with hand-crafted-feature systems to quantify accuracy gains, image quality changes resistance, and computation trade-offs. Meanwhile, hybrid strategies emerge—some work combines traditional methods with characteristics gleaned from CNNs, and some architecture optimizes by bringing domain knowledge from traditional methods. This two-way strategy attempts to take the best from both methods: hand-crafted feature interpretability and efficiency, and deep learning representational ability.

Despite the improvements in finger vein recognition, a variety of issues continue to require full resolution. These include improving the quality and stability of images being taken, improved recognition accuracy under environmental and user variability, and scaling systems well to accommodate large numbers of people [4]. Finger vein recognition has evolved from a creative concept to a complete biometric technology with numerous real-world applications. There needs to be ongoing research to overcome its current limitations and determine its complete potential in secure and convenient identity authentication systems.

1.2 Finger Vein Image Acquisition Methods

The finger vein image acquisition is an essential module of biometric systems that employs near-infrared (NIR) light (typically 700–1000 nm) to digitalize the unique subcutaneous vein patterns in a finger. Some examples of devices used for finger vein image acquisition are shown in Figure 1.2.



Figure 1.2: Examples of Image Acquisition Devices for Finger Vein Recognition

Four acquisition modes are extensively used: light reflection, light transmission, side illumination, and bottom light transmission. They differ primarily in the direction of the NIR light and the image sensor relative to the finger. In the technique of light reflection, as shown in , the infrared light is placed in line with the image sensor and the finger is put in front. The technique is of low power consumption, low price, and

can be implemented in miniature devices. However, since the infrared light only passes through the skin approximately 1 mm deep, the image contrast of the captured image tends to be poor, and high-end infrared sources must be employed. High-resolution image processing is also necessary to achieve salient features from low-contrast or thin veins.

The light transmission technique positions the infrared light source across from the image sensor with the finger in between. This arrangement provides clear contrast images since the infrared light travels through the whole finger. However, image quality varies considerably between areas because of finger thickness and inner structure, leading to inhomogeneity and decreased reliability over various samples.

Side lighting is the procedure of placing infrared lights to the sides of the finger. This procedure allows more scattering of light through the tissue, which enhances image sharpness and contrast against the reflection of light. It is more complex and costly to implement due to the fact that it involves multiple components and a larger design.

The bottom transmission of light method addresses portability in the sense that both the infrared sensor and light are placed underneath the finger. It offers high-definition images similar to side lighting images but with much cost and size in excess for handheld application. The method combines sharp image gathering with small and easy hardware configuration.

The illustrations of these techniques are shown in Figures 1.3, 1.4, 1.5, and 1.6, adapted from the work of Kolivand et al. [5].

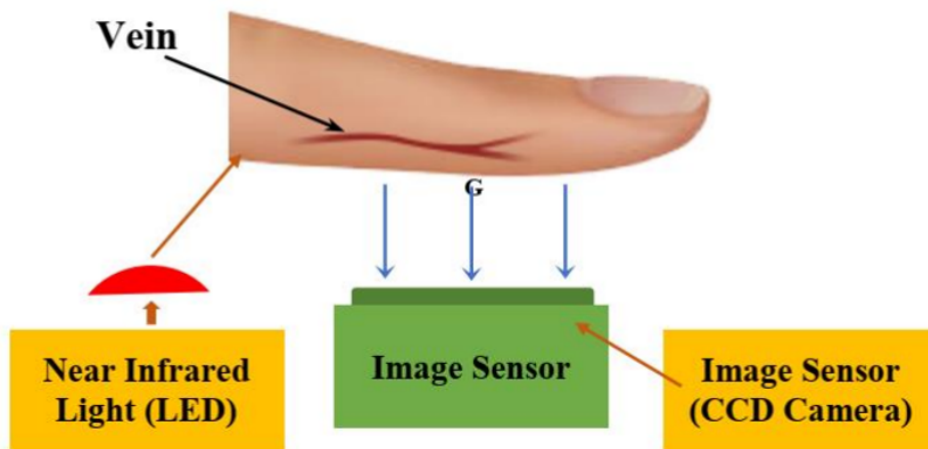


Figure 1.3: Light Reflection Technique [5]

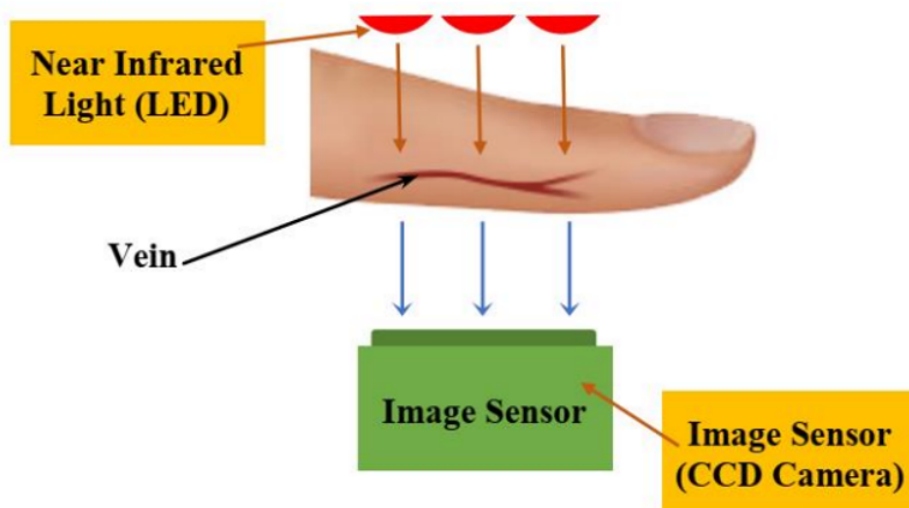


Figure 1.4: Light Transmission Technique [5]

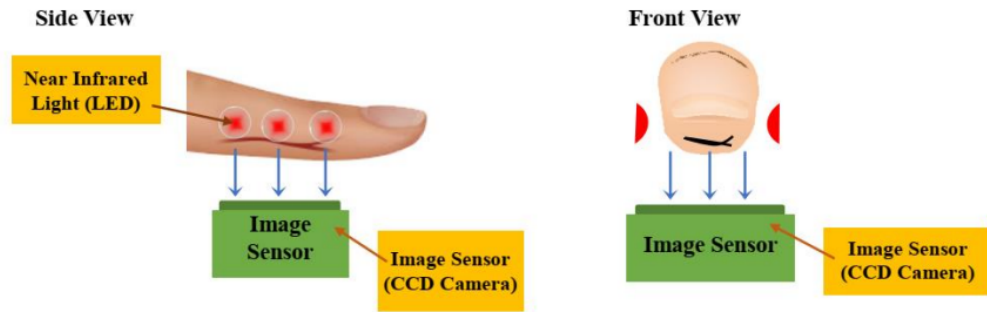


Figure 1.5: Side Lighting Technique [5]

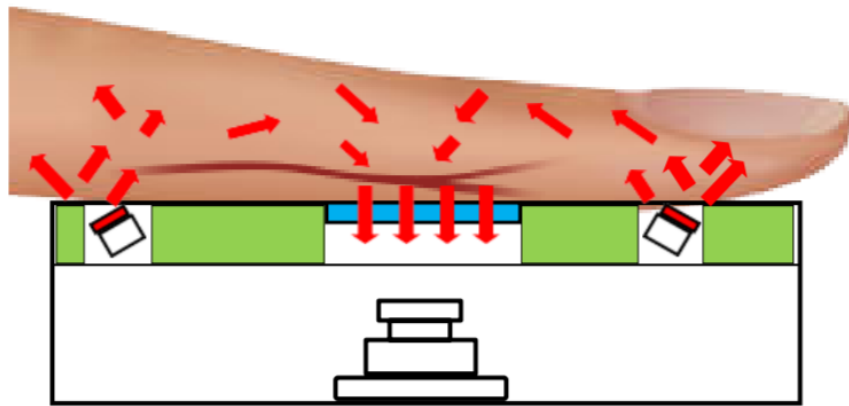


Figure 1.6: Bottom Transmission of Light Technique [5]

Some variables such as finger thickness, fat tissue, coloration of the epidermis, orientation of the finger, surrounding illumination, and image acquisition system quality may also affect sharpness of images. Poor acquisition conditions tend to produce blurry, skewed, dark, or too intense images, which are not good for identification performance and increase preprocessing and feature extraction complexity. Proper finger vein image acquisition remains the basis of secure and accurate biometric identification despite these limitations.

1.3 Preprocessing in Finger Vein Recognition Systems

In finger vein recognition, the quality of the raw image strongly influences every subsequent stage of the pipeline, which is why preprocessing is considered a critical step. This process is illustrated in Figure 1.7. The initial normalization step improves the visibility of vein patterns by adjusting brightness and contrast so that features are represented more consistently across images. Subsequent noise-reduction operations suppress distortions such as Gaussian or impulse noise while, as far as possible, preserving the fine structure of the veins. A selection of normalization and denoising techniques frequently reported in the literature is summarized in Figure 1.8.

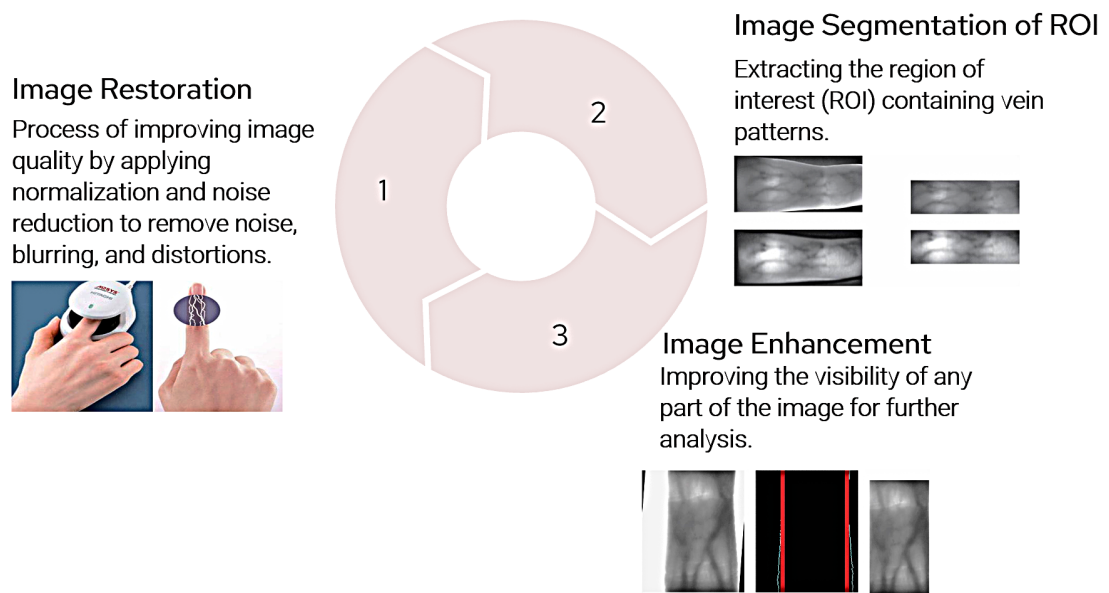


Figure 1.7: Preprocessing Pipeline for Finger Vein Image Enhancement

Normalization	Noise Reduction	Composite Noise Reduction	Hybrid Methods
Min-Max Scaling	Median Filtering	Median + Winner filter + Niblack method	NSDFB + Frangi filtering
Histogram Stretching	Bilateral Filtering	Image layer separation (ILS), Gaussian blur	Fusion of Gabor filter and Retinex filter
Gamma Correction	Non-Local Means (NLM) Filtering		Elliptic high pass filter
Z-Score Normalization	Wavelet Denoising		Gabor filter and Canny edge detector
Logarithmic Transformation	Anisotropic Diffusion Filtering		
Histogram Equalization	Total Variation Denoising (TV)		
CLAHE	Gaussian Filtering		
Retinex filter	Modified Gaussian high-pass filter		
Support Vector Regression (SVR)	Multiscale matched filtering		
Interval type-2 fuzzy set			

Figure 1.8: Normalization and Noise Reduction Methods for Finger Vein Images

After normalization and noise reduction, the next step is Region-of-Interest (ROI) segmentation. Here the goal is to keep only the vein region that is useful for identification and discard background pixels that carry no biometric information. A well-chosen ROI reduces the amount of data the system must process and usually leads to better matching performance. In practice, however, defining a stable ROI is difficult when fingers are not placed consistently—when they are rotated, shifted, or slightly skewed on the sensor.

Several approaches have been proposed to manage these inconsistencies. One of the simplest is fixed-window cropping, but this method is easily disrupted by even slight shifts in finger placement. Other techniques try to locate the finger using anatomical landmarks such as edges or joints; however, their performance tends to drop when the finger is bent or positioned at an angle. Edge-based methods combined with rotational alignment offer better tolerance to orientation changes, yet they often capture more background than intended. This additional clutter reduces the clarity of the ROI and

ultimately affects the reliability of the extracted vein pattern.

1.4 Feature Extraction in Finger Vein Identification Systems

Once the image has been normalized, denoised, and the region of interest selected, the system moves on to feature extraction and matching. The goal at this point is straightforward: to turn the vein pattern in each finger into a compact description that can be stored, compared with other samples, and ultimately used to decide whether two images belong to the same person.

1.4.1 Vein Pattern-Based Techniques

Vein pattern-based methods are the oldest and most widely used family of approaches in finger vein recognition. They work directly with the vascular layout seen in near-infrared images and usually follow two main steps. First, the vein pattern is highlighted or segmented; then, this extracted structure is compared across different images. Researchers have explored several ways to carry out the first step. Maximum-curvature and recursive line-tracking methods look for valleys in the intensity profile, taking advantage of the fact that veins show up as dark, groove-like regions under NIR illumination. Region-growing methods start from a few seed pixels placed inside a vein and gradually add neighbouring pixels with similar brightness, allowing the region to spread along the vessel while leaving most of the background behind. Approaches based on Gabor filters follow a different idea. They examine the image at multiple spatial frequencies and orientations, which makes elongated, vessel-like structures stand out while attenuating background textures and noise that carry little discriminative value. This filtering stage typically yields a binary representation of the vein network. For matching, two vein maps are registered and their similarity is measured by the extent of pixel-level overlap, usually with a small tolerance to account for minor finger translations and slight rotations. Despite being straightforward and easy to interpret, these approaches are highly dependent on

segmentation quality; consequently, their reliability hinges on accurate and consistent region-of-interest (ROI) extraction.

1.4.2 Shape Descriptor Methods

Shape-based descriptors characterize a vein pattern by its structural form rather than by raw intensity values. The central idea is to summarize the extracted vasculature through compact geometric and topological cues that are well suited for matching. A typical first step is skeletonization, which reduces the segmented vein region to a one-pixel-wide centerline. This representation makes it natural to model the vein network as a graph consisting of branches and junctions. This graph representation allows matching algorithms to compare characteristics such as branching points, endpoints, and path lengths. Other descriptors work directly with the vein contour. For instance, Fourier descriptors represent the vein contour using frequency coefficients, which lets the system compare shapes without being much affected by rotation or small deformations. Shape-context descriptors take a different route by focusing on where vein points lie relative to one another. They encode this arrangement using reference bins, so each point is described in relation to its local neighborhood. When segmentation is strong, this representation provides an efficient geometric signature for matching. The downside is that the method is unforgiving to imperfect extraction; gaps in the vein map or extra branches created during skeletonization can reduce performance.

1.4.3 Texture-Based Approaches

Texture-based approaches interpret finger-vein images as textured patterns rather than attempting to explicitly delineate vessels as geometric objects. Instead of extracting vessel boundaries, they characterize local intensity variations through statistical descriptions computed over small neighborhoods. Local Binary Patterns (LBP) and its variants are among the most widely used examples: each pixel is compared with its

surrounding neighbors and the outcome is encoded as a short binary pattern. Because this encoding depends on relative differences rather than absolute brightness, it tends to remain stable under gradual illumination changes.

Other descriptors capture texture at a broader scale. For example, the Gray-Level Co-occurrence Matrix (GLCM) summarizes how frequently pairs of intensity levels occur in close proximity and derives measures such as contrast and homogeneity from these statistics. Local Directional Code (LDC), in contrast, focuses on gradient orientations, thereby strengthening directional cues associated with long and thin vascular structures.

A key practical benefit of these texture-based methods is that they can operate directly on the ROI image, without requiring a perfectly segmented vein map. This makes them attractive when images are noisy, low-contrast, or when segmentation is unreliable. The trade-off, however, is that treating the vein pattern primarily as texture may under-represent fine structural information that shape- or topology-driven methods are designed to preserve.

1.4.4 Dimensionality-Reduction Based Methods

Dimensionality reduction is widely used in finger-vein recognition for a simple reason: both images and many extracted feature representations can become very high-dimensional, which makes storage inefficient and matching slow. In practice, keeping every dimension is rarely necessary. A more effective approach is to project the data into a lower-dimensional space that preserves the variations that help distinguish identities, while filtering out components that mainly reflect redundancy or noise. Among classical techniques, Principal Component Analysis (PCA) is commonly adopted to obtain a compact representation, whereas Linear Discriminant

Analysis (LDA) is often used when improved class separability is the main objective. PCA finds the directions along which the data vary most, while LDA takes a different approach. LDA is built for classification, so it tries to bring samples from the same class closer together and push samples from different classes farther apart. More recent approaches use manifold-learning ideas such as Isomap or Locally Linear Embedding (LLE). These aim to preserve the curved, nonlinear relationships in the data rather than relying purely on overall variance.

In finger vein recognition, dimensionality-reduction methods are often applied on top of existing descriptors, for example by running PCA on LBP histograms or shape-based features, to shrink storage requirements, accelerate matching, and increase robustness to noise. Despite these benefits, there is a notable practical limitation: the learned projection is typically tuned for a fixed set of enrolled users. When new users join the system, the projection space may need to be recomputed or at least updated, which makes large-scale or rapidly changing deployments harder to maintain.

1.4.5 Deep Learning-Based Solutions (CNNs)

Convolutional neural networks (CNNs) are now widely used as end-to-end models for finger-vein recognition. Instead of relying on hand-crafted descriptors, they learn several layers of features directly from the raw images. Well-known architectures such as AlexNet, ResNet, and DenseNet can pick up vein patterns at different spatial scales through their stacked convolutional blocks, and attention modules (for example, squeeze-and-excitation units) help the network focus on the most informative vascular regions. For matching, Siamese CNNs learn a notion of similarity between image pairs, while hybrid designs, such as Gabor-CNN models, combine classical filter banks with learned representations. These approaches remove

much of the manual feature-engineering effort but come with clear costs: they typically need large, carefully annotated datasets and considerable computing power [6]. In addition, because their internal decision process is difficult to interpret, they are often treated as “black boxes,” which contrasts with the more transparent behaviour of rule-based methods.

1.4.6 Hybrid Methods

Hybrid methods combine complementary feature-extraction and learning strategies so that limitations of one component are mitigated by the strengths of another. One common design pairs hand-crafted texture descriptors such as LBP with PCA, reducing the dimensionality of the feature space and making subsequent matching more efficient. Another direction first constructs curvature-based vein maps and then feeds them into a CNN, allowing the network to learn from a cleaner and more structured representation of the vasculature. In many studies, such hybrid pipelines achieve highly competitive performance because they preserve the interpretability and stability of classical descriptors while also benefiting from the representation learning capability of deep models. The main disadvantage is increased system complexity and a larger number of hyperparameters, which can make deployment and long-term maintenance more demanding.

1.5 Matching in Finger Vein Recognition Systems

The matching stage acts as the final decision component in a finger-vein recognition system. Once features have been extracted, this stage evaluates how similar two samples are and decides whether they correspond to the same individual. The choice of similarity measure depends strongly on the representation of the features. For binary representations such as a vein mask, Hamming distance is a natural choice because it counts the bit locations where two patterns differ. When the descriptors are numerical Manhattan distance is often used instead, since it is typically less affected

by small fluctuations and outliers.

For high-dimensional descriptors or deep feature embeddings produced by convolutional neural networks, Euclidean distance is frequently used to measure proximity in the embedding space. Cosine similarity provides an alternative by focusing on the angular relationship between feature vectors, making it largely independent of their magnitude. In some systems, these similarity measures are further combined with machine-learning classifiers, most notably support vector machines, which learn a decision boundary that separates genuine matches from non-matches and improves the reliability of the final accept–reject decision.

1.6 Research Gap

Finger vein recognition has grown through a variety of feature-extraction strategies, ranging from vein-pattern analysis and shape-based descriptors to texture-oriented methods such as LBP. Systems that rely on LBP usually handle vein textures well at a single spatial scale, but most versions use only one radius. This limitation can make the method less stable when lighting changes, when fingers shift in orientation, or when images are captured in different sessions. Deep learning approaches, particularly CNNs, have also produced strong results in finger vein identification. Their performance, however, often depends on access to large, well-balanced datasets. When the available data is limited, CNN models tend to overfit and their ability to generalize drops noticeably [7]. hand-crafted descriptors, on the other hand, continue to perform reliably under small-data conditions, remain computationally lightweight, and are easier to interpret. Motivated by these advantages, this work introduces a fused multi-radius version of LBP^{riu2}. Despite the progress in the field, there has been no systematic investigation of whether multi-radius fused LBP^{riu2} features can be further improved by combining them with dimensionality-reduction techniques such

as PCA, 2DPCA, or $(2D)^2$ PCA. It is also not well understood how these combinations compare with standard single-radius descriptors or with recent CNN-based models when tested across different datasets and evaluation protocols.

1.7 Chapters and Sections

The remainder of this thesis is organized as follows. Chapter 2 reviews the related literature with a particular emphasis on feature extraction methods. Chapter 3 introduces the datasets and experimental methodology, describing the public databases employed (FV-USM, MMCBNU-6000, and UTFVP), the protocol splits, and the two operational strategies designed for systematic benchmarking. Chapter 4 presents the proposed finger-vein identification framework in detail, covering the ROI segmentation and preprocessing pipeline, as well as the implemented feature extraction techniques (namely PCA, 2DPCA, $(2D)^2$ PCA, rotation-invariant LBP, and the multi-radius fused $\text{LBP}_{(8,1)(16,1)(8,2)}^{\text{riu}2}$). It also explores their dimensionality-reduced variants using PCA, 2DPCA, and $(2D)^2$ PCA, in addition to the adapted AlexNet architecture with its training and deployment considerations under the two strategies. Chapter 5 reports and analyzes the identification results through quantitative tables, CMC curves, rank-based metrics, and comparative evaluations with existing methods, along with discussions on dataset- and protocol-dependent behaviors. Finally, Chapter 6 concludes the thesis and outlines directions for future research.

Chapter 2

LITERATURE REVIEW

Finger vein recognition did not become reliable overnight. Early systems were built around a simple idea: extract vein lines from near-infrared images and then match those extracted patterns across samples. When segmentation worked well, the results could look convincing; in real captures, however, the pipeline was often fragile. Small changes in image quality, uneven illumination, or slight shifts in finger placement could distort the extracted structure, and those errors would show up immediately during matching. Instead of betting everything on a clean vein map, later work started to favor features that still behave sensibly when the input is messy. Texture descriptors fit that need well, because they summarize local intensity structure even when the vein boundaries are not perfectly separated from the background. LBP became one of the most common choices, mainly because it is lightweight to compute and tends to remain stable under mild illumination changes; rotation-invariant variants further helped when the finger is not placed in exactly the same orientation.

In parallel, dimensionality-reduction techniques were used to keep representations compact and matching fast as databases grew. More recently, deep learning models have been explored for learning representations directly from data and have achieved strong performance in some settings. Nevertheless, classical and hybrid pipelines remain widely used in finger vein recognition, largely because many public datasets are not large enough to support extensive training without overfitting. Across this body of work, one recurring insight stands out: scale matters, and single-radius

descriptors can be unstable across sessions, which helps explain the continued interest in multi-scale and multi-radius designs.

2.1 Background of the Field and Related Works

Early studies in finger vein recognition mainly focused on extracting visible vein lines from pre-segmented images. The idea was simple, yet the approach was fragile since any segmentation error tended to propagate into the matching stage. As researchers explored alternatives that required less preprocessing, Local Binary Patterns and other texture descriptors began to draw attention. LBP had an obvious advantage since it encoded local texture directly from the raw image and eliminated the need for explicit vein segmentation. Originally proposed by Ojala et al. [8] for texture classification, LBP transforms each neighborhood into a binary pattern by comparing the intensities of pixels. A later version with rotation invariance [9] contributed to reducing the operator's sensitivity to finger rotation.

One of the earliest attempts to apply LBP to finger vein analysis was by Lee et al. [10], where minutiae alignment was combined with LBP coding to form a binary representation of the vein. The false rejection rate was significantly reduced in their system. Following this, H. C. Lee et al. [11] developed a weighted LBP approach with an SVM classifier. They weighted the LBP features according to the amount of vein information contained in each region, rather than treating every region equally. This weighting strategy turned out to be more effective and even outperformed traditional LBP and several vein-based classical approaches.

As finger-vein recognition gained attention, several researchers began to adapt the Local Binary Pattern (LBP) framework so that it better reflects the structure of vein patterns rather than generic texture. One of the early contributions in this direction

was reported by Rosdi et al. [12], who introduced Local Line Binary Pattern (LLBP). The main change is straightforward: instead of using the usual two-dimensional neighborhood, LLBP samples along a one-dimensional configuration. This makes the descriptor more sensitive to the dominant local orientation of vein lines. To better capture elongated structures, a vertical version of LLBP was later introduced as well. Around the same period, Yang et al. [13] introduced the Personalized Best Bit Map (PBBM). Rather than changing the sampling pattern itself, PBBM asks a more practical question: which LBP bits remain consistent for a given user across multiple samples? Bits that fluctuate noticeably from one capture to another are treated as unreliable and are removed. In general, keeping only these stable positions improves robustness and produces more repeatable recognition results.

Later research shifted from the early LBP concepts toward improving the descriptive power of LBP-based descriptors. Lu et al. [14] proposed the Generalized Local Line Binary Pattern (GLLBP), which extends LLBP by supporting multiple sampling orientations instead of committing to a single fixed direction. This matters in real data because vein patterns are not perfectly straight; they can appear curved or tilted, and small pose changes or acquisition variability can make the dominant direction vary across the image.

Xie et al. [15] refined the directional modeling further through the Local Opposite Directional Pattern (LODP). Unlike approaches that rely only on standard local intensity comparisons, LODP explicitly models intensity transitions along opposite directions around each pixel. By encoding these directional changes, it tends to give a more dependable local texture characterization, particularly when contrast varies across the image or illumination is not uniform.

Besides descriptive accuracy, computational cost has also been a recurring concern in finger-vein recognition, especially for practical systems. Liu et al. [16] addressed this by proposing an efficient LBP-based scheme that integrates a Random Forest regressor to estimate vein-region intensities. The goal was to accelerate feature extraction while keeping recognition performance essentially unchanged. Finally, Shaheed et al. [17] presented a comprehensive survey of finger-vein recognition methods and emphasized that LBP and its many variants remain widely used hand-crafted descriptors in this field, mainly because they are simple to compute, efficient, and often deliver reliably competitive performance.

Hybrid approaches gradually attracted attention as well. Hu et al. [18] combined rotation-invariant uniform LBP with 2DPCA and found steady improvements with the two used together. Their subsequent work further introduced Multi-scale Uniform LBP (MULBP) [19], in which uniform LBP histograms were computed over multiple spatial scales after dividing the image into blocks. Tests on public datasets reported recognition rates higher than 99 percent. Extending this line of effort, Zhang et al. [20] proposed Multi-Feature Partitioned LBP (MFPLBP), in which several regions are partitioned to cope with global shapes and fine local textures effectively. MFPLBP outperforms traditional LBP, PLBP, and MULBP. More recently, the review by Kolivand et al. [5] reaffirmed the central role of LBP-based descriptors even while CNN-driven and hybrid models gained broader influence.

Classical dimensionality-reduction techniques also played a central role in many pipelines. PCA is often used because it preserves most of the variance in the original data [21]. 2DPCA being a matrix-based extension of PCA [22] operates directly on image matrices instead of vectorized images. This allows spatial structure to be

retained an advantage for vein pattern analysis. Several variants appeared over time. Two-directional 2DPCA (2D2PCA) [23] and Relaxed 2DPCA (R2DPCA) [24] incorporate sparsity and class-label information into the projection, increasing the discriminative power of the extracted features.

Deep learning models such as AlexNet [25] introduced a different approach by learning features directly from raw images, removing the need to design descriptors by hand. Comparative studies consistently show that hand-crafted methods tend to perform well in situations where datasets are small or difficult to label. This is because they do not rely on large-scale training data. Conversely, CNNs tend to reach their full potential only when extensive and carefully curated datasets are available. Currently, much of the ongoing progress in finger vein recognition research is shaped by the continuous interplay between classical descriptors and modern deep learning models.

Finger vein recognition research has evolved through several distinct phases rather than following a single linear trajectory. Early systems depended heavily on segmentation, but later studies increasingly moved toward texture operators such as LBP and, afterward, multi-scale variants designed to capture patterns at different spatial resolutions. Dimensionality-reduction techniques then became common building blocks in many pipelines, and more recently deep learning models have introduced feature learning directly from data. Even with this steady evolution, many LBP-based systems still adopt only one sampling radius. In practice, a single radius can be restrictive, because it often increases sensitivity to illumination changes, differences in finger posture, and natural irregularities in vein appearance. These limitations help explain the growing interest in multi-radius fused $\text{LBP}^{\text{riu}2}$ descriptors, which aggregate information across multiple scales and typically provide a richer and

more discriminative representation. At the same time, there has been increasing attention to how such multi-radius features behave when paired with modern dimensionality-reduction methods, particularly under larger and more diverse evaluation settings.

In recent years, several new trends have emerged in finger vein recognition. Table ?? summarizes a selection of notable contributions between 2020 and 2025, along with the methods, datasets, and reported performance of each.

Table 2.1: Overview of Major Finger Vein Recognition Contributions (2020–2025)

Ref.	Method	Dataset	Result
[26]	A deep cascade correlation	SDUMLA-HMT	94% accuracy, 96% F-score,
	model with Gabor filters and		66 ms speed, 55 MB model
	bipolar activation		size. Robust and scalable
[27]			Gabor features.
	3DFV/SNet:	SCUT-LFMB-	2.6–2.8% EER. Real-time,
	Silhouette-based 3D finger	3DPV-FV;	rotation-invariant 3D
	vein reconstruction using	SCUT-3DFV-V1	recognition outperforming
	epipolar constraints and		prior methods.
	rotation-invariant 3D CNN		

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Table 2.1 – continued from previous page

Ref.	Method	Dataset	Result
[28]	FedFV: Federated learning	MMCBNU-6000;	0.96% EER; 98.79%
	with MobileNetV2 using	HKPU-FV;	TAR@FAR=0.01 across
	personalized layers and	PLUSVein-FV;	nine datasets.
	FedWPR for non-IID data	SDUMLA-HMT;	Privacy-preserving and
		THU-FVFDt;	computation-efficient.
		FV-USM; UTFVP;	
		VERA; SCUT-FV	

Continued on next page

Table 2.1 – continued from previous page

Ref.	Method	Dataset	Result
[29]	LPQ + LDP feature	HKPU;	~98% accuracy, 0.10%
	extraction with	SDUMLA-HMT;	EER, faster runtime than
	GWO-optimized SVM	FV-USM;	classical SVM. Robust to
	classification	THU-FVFDT2	noise and illumination changes.
[30]	Gabor feature extraction	FV-USM; PolyU;	Over 96% accuracy with low
	with NGA-Clearing feature	SDUMLA; UTFVP	EER. Outperforms PCA and
	selection and KNN classifier		GA; suitable for IoT environments.

Continued on next page

Table 2.1 – continued from previous page

Ref.	Method	Dataset	Result
[31]	Preprocessing via bilateral	SDUMLA-HMT;	99.94% recognition rate
	filtering and contour-based	SCUT-FVD; VERA	using adaptive Gabor +
	ROI; feature extraction using		SURF. GPU-accelerated and
	SIFT and SURF		training-efficient.
[32]	Hybrid hand-crafted features	PolyU; NTU;	98.89–100% accuracy for
	(PWBDC, Gabor, Curvature,	HKPU;	hand-crafted methods; DBN
	WLD, LBP) combined with	SDUMLA-FV;	achieves 0.76% EER with
	DBN	PKU; FV-TJ;	improved noise robustness.
		FV-USM	

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Table 2.1 – continued from previous page

Ref.	Method	Dataset	Result
[33]	Dynamic-Softmax with	FV-USM;	State-of-the-art performance
	adaptive margin loss based	HKPU-FV;	(0.31–1.59% EER) via
	on posterior probabilities	SDUMLA-HMT;	adaptive margins and cosine
		MMCBNU-6000;	similarity optimization.
[34]		THU-FVFDT3	
	Cancelable templates using	SDUMLA-HMT;	High accuracy with privacy
	Gaussian random projection	Chimeric dataset	compliance. Resistant to
	and hash-based encoding	(LFW + UTFVP)	pre-image attacks; suitable for real-time authentication.

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Table 2.1 – continued from previous page

Ref.	Method	Dataset	Result
[35]	Enhanced Elman Spike	SDUMLA-HMT;	Accuracy, sensitivity, and
	Neural Network optimized	CASIA V5	precision improved by
	via GSO with LECF and		20–24%. Robust for
	ACEWT preprocessing		transaction authentication.

Continued on next page

Table 2.1 – continued from previous page

Ref.	Method	Dataset	Result
[5]	Comprehensive review of	Hitachi Res. Lab.;	90–99.67% accuracy; EER
	preprocessing, feature	Int. Biom. Group;	as low as 0.0038%.
	extraction, and classification	Hitachi-Kyushu;	Highlights CNN and Gabor
	techniques	PKU v2–v4;	dominance and dataset
		GUC45;	impact.
		SDUMLA-HMT;	
		HKPU; UTFVP;	
		MMCBNU-6000;	
		CFVD; Shandong	
		Univ.; FV-USM	

Chapter 3

DATASETS AND EVALUATION METHODOLOGY

The performance of a finger vein recognition system cannot be judged in isolation from the data and evaluation procedures used to test it. Variations in acquisition hardware, capture conditions, subject populations, and protocol design can all influence reported recognition accuracy, sometimes quite substantially. For this reason, experimental studies in biometric research rely heavily on publicly available datasets and carefully defined evaluation strategies that allow results to be interpreted and compared in a meaningful way. This chapter outlines the experimental setup used throughout the thesis, including the datasets and the validation procedure. It begins by introducing three widely used finger-vein datasets (FV-USM, MMCBNU-6000, and UTFVP) and highlighting their structure and relevance for evaluating recognition algorithms. The chapter then details the experimental protocols and processing steps adopted to ensure that comparisons remain consistent, reproducible, and informative across methods and datasets.

3.1 Finger Vein Datasets for Experimental Evaluation

Finger-vein recognition relies on the distinctive structure of subcutaneous vascular patterns, which are not easily duplicated. To support fair comparisons between methods, evaluation is typically conducted on public datasets with clearly defined acquisition settings. In this section, FV-USM, MMCBNU-6000, and UTFVP are presented, with a focus on how they were collected and what characteristics matter for experimental benchmarking.

3.1.1 FV-USM

The FV-USM dataset [36] contains 5,904 grayscale images acquired from 123 subjects (83 males and 40 females). For each subject, images were captured from four fingers (left/right index and left/right middle), yielding four finger classes per person. Data collection was performed over two acquisition sessions, with six samples recorded per finger in each session, resulting in a total of $123 \times 4 \times 6 \times 2 = 5,904$ images (see examples in Figure 3.1). The dataset also provides pre-extracted regions of interest (ROIs) of size 300×100 pixels, cropped from the original 640×480 images, as shown in Figure 3.2. All images are stored as 8-bit grayscale JPEG files, which offers a practical balance between storage efficiency and preservation of fine vein-texture information needed for recognition.

Since FV-USM contains two capture sessions per subject, we adopt a cross-session evaluation protocol in which enrollment (training) and probe (testing) images are sourced from both sessions. In this setting, genuine matches are defined as images of the same finger captured in the *same* session, allowing recognition performance to be assessed while controlling for inter-session variations such as hand positioning, illumination conditions, and sensor drift.

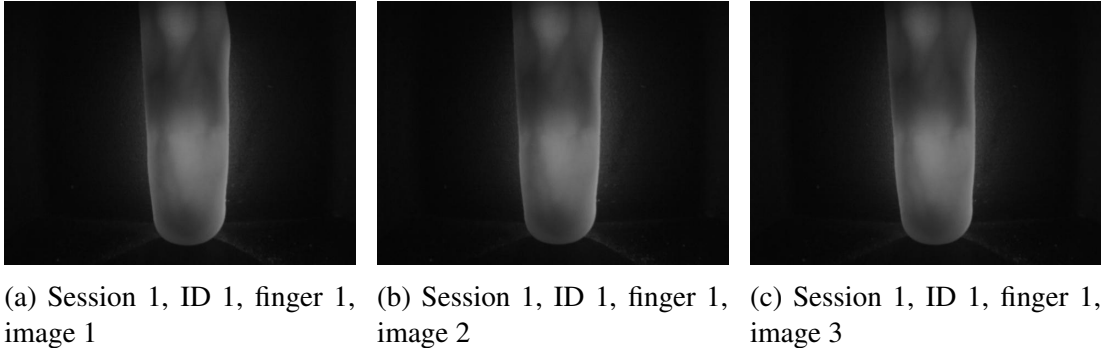
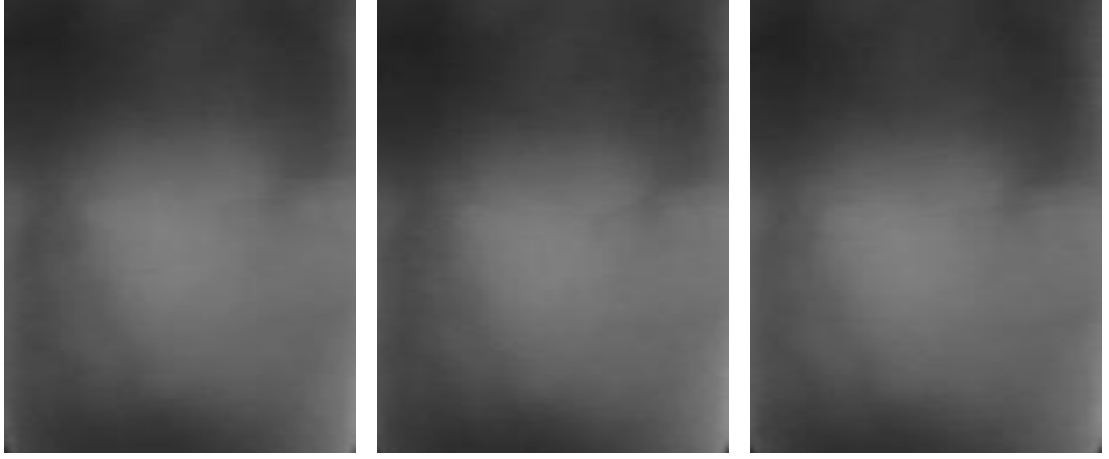


Figure 3.1: Original Captured Images in FV-USM dataset

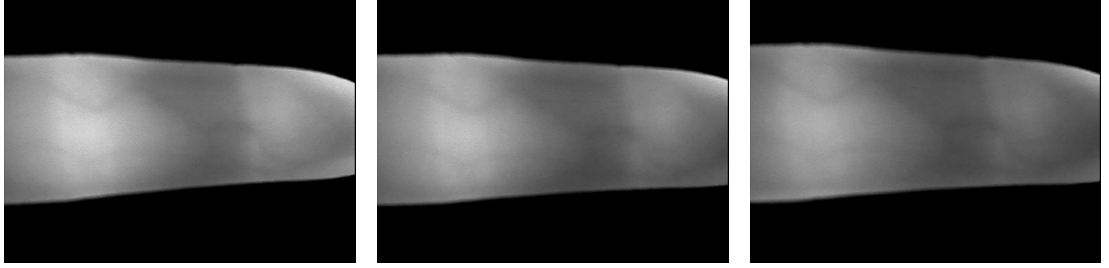


(a) Session 1, ID 1, finger 1, image 1 (ROI) (b) Session 1, ID 1, finger 1, image 2 (ROI) (c) Session 1, ID 1, finger 1, image 3 (ROI)

Figure 3.2: Extracted ROI Images in FV-USM Dataset

3.1.2 MMCBNU-6000

The MMCBNU-6000 database consists of 6,000 images of 100 individuals (83 males and 17 females) with 20 nationalities in the age bracket of 16-72 years. The images were all captured during a single session as compared to FV-USM. Six fingers (right and left index, middle, and ring fingers) per individual were contributed with 10 samples on each finger, resulting in 6,000 images. They are raw uncompressed BMP images shot at a resolution of 640×480 pixels in 8-bit grayscale [37]. Some examples of these original captured images are provided in Figure 3.3. The data set is already ready to be used in experimental testing as it comes with pre-extracted regions of interest (ROIs), which are usually 128×60 pixels [38]. Some examples of these ROI images are shown in Figure 3.4. Its size and demographic mix make it a practical benchmark without compromising natural finger placement variations.



(a) ID 1, left index finger, image 1 (b) ID 1, left index finger, image 2 (c) ID 1, left index finger, image 3

Figure 3.3: Original Images from MMCBNU-6000 Dataset



(a) ID 1, ROI of left index finger, image 1 (b) ID 1, ROI of left index finger, image 2 (c) ID 1, ROI of left index finger, image 3

Figure 3.4: ROI Images from MMCBNU-6000 Dataset

3.1.3 UTFVP

The UTFVP database [39] contains 1,440 grayscale images ($60 \text{ volunteers} \times 6 \text{ fingers per volunteer} \times 4 \text{ samples per finger}$) acquired in two capture sessions. The original images have a resolution of 672×380 pixels and are distributed as 8-bit PNG files. Although the images were captured in two sessions, the publicly released UTFVP collection is provided as a single, undifferentiated set (i.e., without explicit session labels), so the dataset is effectively treated as a single-session corpus unless session information is reconstructed from auxiliary metadata. UTFVP does not include pre-extracted ROIs; therefore, ROIs must be extracted prior to feature computation or matching. In our preprocessing pipeline each image is passed through ROI extraction and each resulting ROI is augmented with three variant transforms. This augmentation policy expands each original ROI into four variants; given the four original samples per finger in UTFVP, the prepared dataset contains $4 \times 4 = 16$ ROIs per finger instance, improving intra-class variability and facilitating stable protocol

splits.

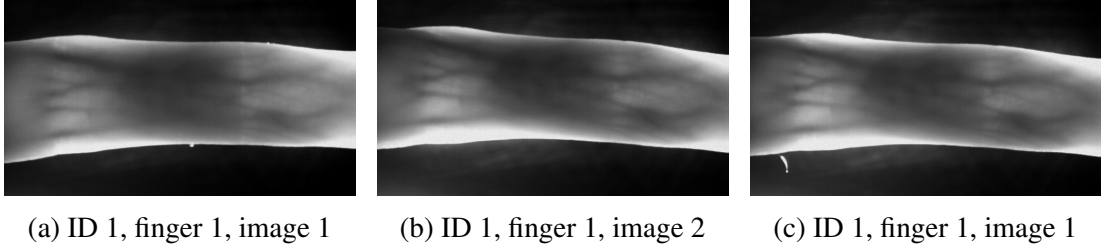


Figure 3.5: Original Images from UTFVP Dataset

3.2 Experimental Protocols and Validation Strategies

Experimental protocols define how datasets are partitioned into training and testing subsets so that the performance of biometric systems can be evaluated reliably and reproducibly. These partitions measure a system’s generalisation under different conditions (e.g., varying enrollment sizes or session discrepancies). Processing strategies determine how features—such as finger-vein feature maps—are handled: either fused across multiple samples to produce a composite template, or treated independently per sample. This section describes the protocols, the two processing strategies considered, and dataset preparation steps.

3.2.1 Protocols

Three sample-partition protocols are used. For each finger instance, we use:

- Protocol 1: 70% of the samples for training, remainder for testing.
- Protocol 2: 50% of the samples for training, remainder for testing.
- Protocol 3: 90% of the samples for training, remainder for testing.

When a percentage yields a non-integer training count, the training set size is rounded down to the nearest integer and the remaining images are assigned to testing; this rule preserves at least one test sample. These protocol implementations are shown in Table 3.1.

Table 3.1: Training and Testing Data Splits by Protocol

Dataset	Protocol	Training Set	Testing Set
FV-USM	Protocol 1 (70-30)	4 samples per instance (both sessions)	2 samples per instance (both sessions)
	Protocol 2 (50-50)	3 samples per instance (both sessions)	3 samples per instance (both sessions)
	Protocol 3 (90-10)	5 samples per instance (both sessions)	1 sample per instance (both sessions)
MMCBNU_6000	Protocol 1 (70-30)	7 samples per instance	3 samples per instance
	Protocol 2 (50-50)	5 samples per instance	5 samples per instance
	Protocol 3 (90-10)	9 samples per instance	1 sample per instance
UTFVP	Protocol 1 (70-30)	11 samples (synthetically augmented images): 3 (Instance 1), 3 (Instance 2), 3 (Instance 3), 2 (Instance 4)	5 samples: third synthetically augmented sample (Instance 4) plus original samples
	Protocol 2 (50-50)	8 samples: 2 from each Instance 1–4	8 samples: third synthetically augmented sample from each instance plus original samples
	Protocol 3 (90-10)	14 samples: [Original plus 3 synthetically augmented (Instance 1–2)] plus [3 synthetically augmented (Instance 3–4)]	Original samples only (Instance 3–4)

3.2.2 Strategies

1. **Strategy 1:** For each subject, several samples are combined to build a single template, which is then used during matching. This approach acknowledges that individual acquisitions may differ slightly because of noise or minor changes during capture. By aggregating multiple samples, these variations are reduced, making this strategy more suitable for multi-shot experiments.
2. **Strategy 2:** Each sample is treated as an independent template and matched on its own, without applying any fusion. This setting reflects a single-shot scenario and is mainly used as a baseline for performance comparison.

Chapter 4

FINGER VEIN IDENTIFICATION FRAMEWORK: PREPROCESSING, FEATURE EXTRACTION AND MATCHING

Finger vein identification is highly sensitive to how vein patterns are captured, isolated, and represented, since even small changes in finger placement, illumination, sensor properties, or acquisition sessions can obscure the underlying vascular structure if they are not controlled. For this reason, the recognition pipeline adopted in this thesis is designed as an end-to-end framework in which preprocessing, feature extraction, dimensionality reduction, and matching are tightly connected rather than treated as separate steps.

The framework begins with a region-of-interest (ROI) segmentation pipeline, developed in detail on the University of Twente Finger Vein Pattern (UTFVP) dataset, to make images comparable by standardizing spatial alignment and suppressing irrelevant background content through contrast enhancement, boundary detection, atypical placement handling, centerline refinement, orientation correction, and fixed-size ROI extraction. Because acquisition conditions are never perfectly consistent in practice, the extracted ROIs are then augmented to reflect typical variability, and the same preprocessing routine is applied across FV-USM, MMCBNU-6000, and UTFVP to ensure comparable inputs under both evaluation settings: multi-fusion (Strategy 1) and single-finger matching (Strategy 2). With these

standardized inputs in place, the chapter proceeds to the feature extraction methods evaluated in this work, including Principal Component Analysis (PCA), Two-Dimensional Principal Component Analysis (2DPCA), Bidirectional Two-Dimensional Principal Component Analysis (2D²PCA), Rotation-Invariant Uniform Local Binary Patterns (LBP^{riu2}), the proposed Multi-Radius Fused LBP^{riu2}, and their dimensionality-reduced variants, along with an adapted AlexNet-based convolutional neural network; together, these approaches form the methodological backbone for all subsequent experimental analyses in the thesis. Figure 4.1 provides a high-level overview of this end-to-end pipeline, showing how ROI extraction, augmentation, and standardized preprocessing feed a unified training database that supports rank-1 identification through the subsequent classification stage.

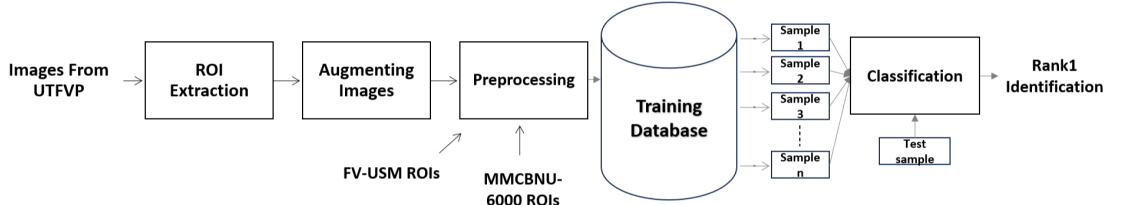


Figure 4.1: Overview Diagram of the ROI Extraction, Augmentation, Preprocessing, and Classification Workflow

4.1 Region of Interest Segmentation for UTFVP Finger Vein Images

The ROI extraction pipeline for finger vein images, which is inspired by the work of Lu et al. [37], consists of six main stages. It is designed to enhance vein visibility, accurately locate finger boundaries, and extract a standardized 60×128 pixel region of interest (see Equation 4.1).

$$\text{Region of Interest Dimensions} = \begin{cases} \text{Height} = 60\text{px} \\ \text{Width} = 128\text{px} \end{cases} \quad (4.1)$$

4.1.1 Contrast Enhancement using CLAHE

The raw finger vein image $I(x,y)$ is first enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE). The image is divided into small tiles of size 8×8 pixels, and each tile's histogram is clipped to avoid over-amplifying noise. The clipped histograms are then equalized and merged using bilinear interpolation.

The histogram of a tile $R_{i,j}$ is calculated as Equation 4.2:

$$H_{i,j}(k) = \sum_{(x,y) \in R_{i,j}} \delta(I(x,y) - k), \quad k \in [0, 255] \quad (4.2)$$

where δ is the Kronecker delta function. The clipped histogram is shown in Equation 4.3:

$$\tilde{H}_{i,j}(k) = \min(H_{i,j}(k), T), \quad T = 0.75 \cdot \frac{64}{256} \quad (4.3)$$

The redistributed histogram is then calculated by Equation 4.4:

$$\hat{H}_{i,j}(k) = \tilde{H}_{i,j}(k) + \frac{\sum_{k=0}^{255} (H_{i,j}(k) - \tilde{H}_{i,j}(k))}{256} \quad (4.4)$$

Here, $H_{i,j}(k)$ is the original histogram of tile $R_{i,j}$, $\tilde{H}_{i,j}(k)$ is the clipped histogram, $\hat{H}_{i,j}(k)$ is the final equalized histogram, and T is the clipping threshold (Equation 4.3).

4.1.2 Edge Detection

To estimate the top and bottom finger boundaries, the image is split into upper and lower halves. Prewitt edge detection is applied to highlight vertical intensity transitions. The convolution kernels are defined as Equation 4.5:

$$G_{\text{upper}} = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad G_{\text{lower}} = -G_{\text{upper}} \quad (4.5)$$

The edge response at each pixel (x,y) is like Equation 4.6:

$$E(x,y) = (I * G)(x,y) \quad (4.6)$$

where I is the image intensity and $*$ denotes 2D convolution.

4.1.3 Abnormal Finger Placement Detection

The variance of the top and bottom edges is evaluated to detect abnormal finger positioning. If high variance is observed in both halves of the finger, vertical padding is added to ensure the finger is fully captured. Mathematically, the condition is shown in Equation 4.7:

$$\text{Abnormal if } \text{Var}(\text{top_edge}[0 : w/2]) > \text{TH} \quad \text{AND} \quad \text{Var}(\text{top_edge}[w/2 : w]) > \text{TH} \quad (4.7)$$

where $\text{Var}(\cdot)$ is the variance function, w is the image width, and $\text{TH} = 38$ is the variance threshold.

4.1.4 Middle Finger Line Cleaning

The central axis of the finger is extracted and cleaned using *DBSCAN clustering* to remove outlier points, as provided in Equation 4.8. A point $\mathbf{p}_i$ is considered a core point if:

$$\|\mathbf{p}_i - \mathbf{p}_j\|_2 < \varepsilon \quad \text{for at least min_samples points} \quad (4.8)$$

where $\mathbf{p}_j$ is any other point in the dataset, $\varepsilon = 10$ pixels is the neighborhood radius, and $\text{min_samples} = 10$. Only inlier points are retained for further processing.

4.1.5 Finger Orientation Estimation

The cleaned middle line is used to estimate the tilt angle θ via linear regression, as shown in Equation 4.9:

$$\theta = \tan^{-1} \left(\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \right) \quad (4.9)$$

where (x_i, y_i) are the coordinates of inlier points, and $(\bar{x}, \bar{y})$ is their mean. Images with $|\theta| > 15^\circ$ are rotated to align the finger vertically.

4.1.6 ROI Extraction

Finally, a region of interest (ROI) of size 60×128 pixels is extracted around the vertical center of the finger. Let x_c denote the horizontal position with maximum vertical intensity. It is calculated with Equation 4.10:

$$x_c = \arg \max_{x \in [W/4, 3W/4]} \sum_{y=0}^{H-1} I(y, x) \quad (4.10)$$

The ROI is then defined with Equation 4.11:

$$\text{ROI} = I[y_c - 30 : y_c + 30, x_c - 64 : x_c + 64] \quad (4.11)$$

where y_c is the vertical center of the finger, and H, W are the image height and width.

4.1.7 Pipeline Summary

The entire ROI extraction pipeline is summarized in Figures 4.2, 4.3, 4.4, and 4.5. ROI extraction is performed during preprocessing; however, saving preprocessed ROIs from the UTFVP dataset requires additional steps (e.g., denoising and histogram equalization), which are discussed in Section 4.2. The final saved ROIs are shown in Figures 4.6, 4.7, and 4.8.

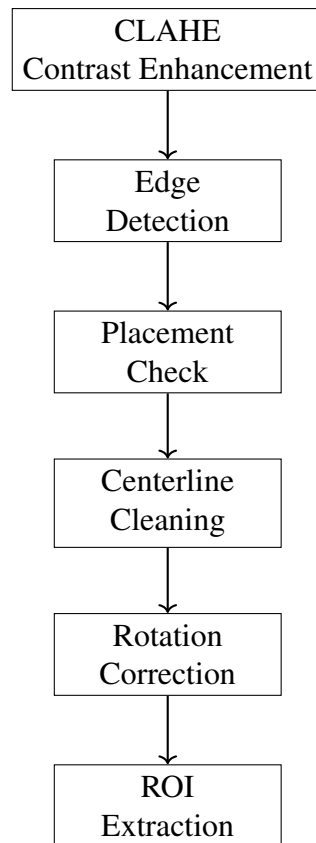


Figure 4.2: Finger Vein ROI Extraction Pipeline in UTFVP Dataset

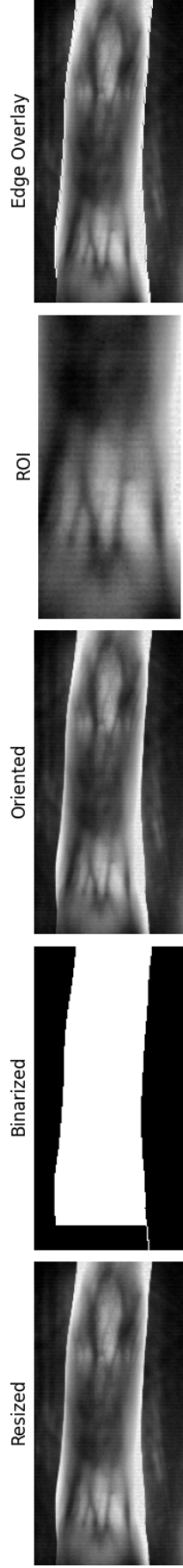


Figure 4.3: ROI Extracted Image from ID 1, Vein 1, and Sample 1 in UTFVP Dataset

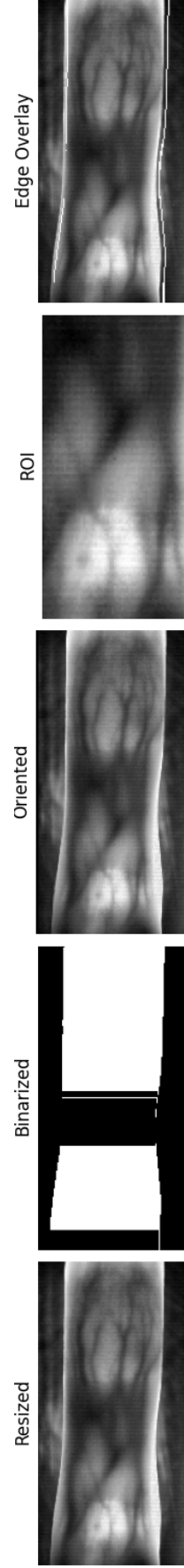


Figure 4.4: ROI Extracted Image from ID 2, Vein 1, and Sample 1 in UTFVP Dataset

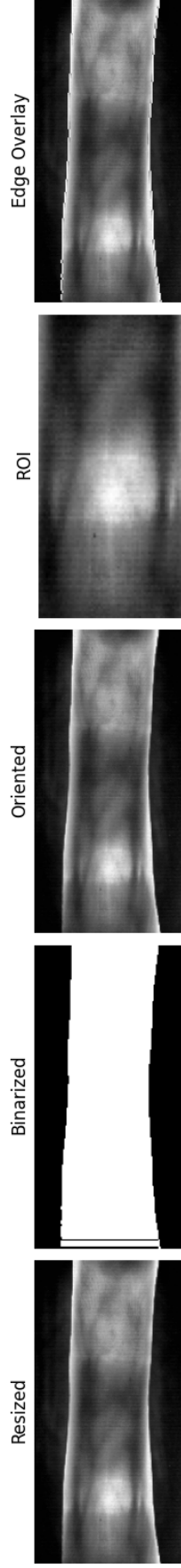


Figure 4.5: ROI Extracted Image from ID 3, Vein 1, and Sample 1 in UTFVP Dataset

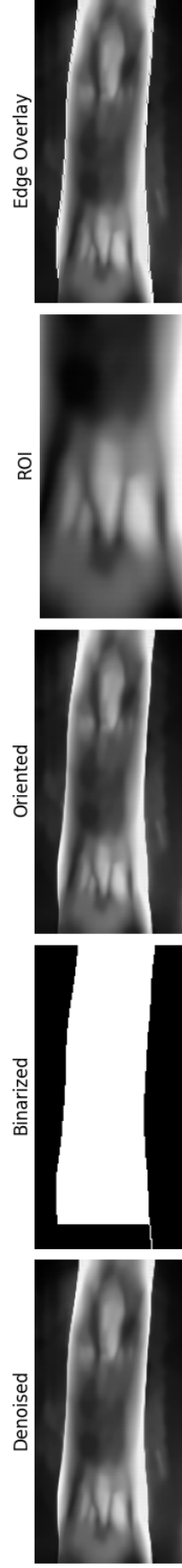


Figure 4.6: ROI Extracted and Preprocessed Image from ID 1, Vein 1, and Sample 1 in UTFVP Dataset

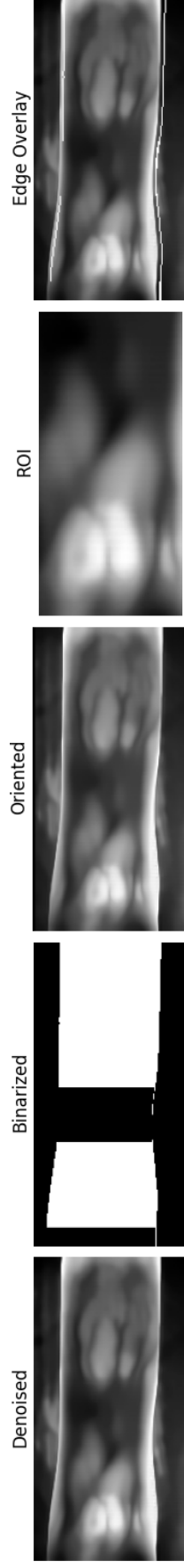


Figure 4.7: ROI Extracted and Preprocessed Image from ID 2, vein 1, and Sample 1 in UTFVP Dataset

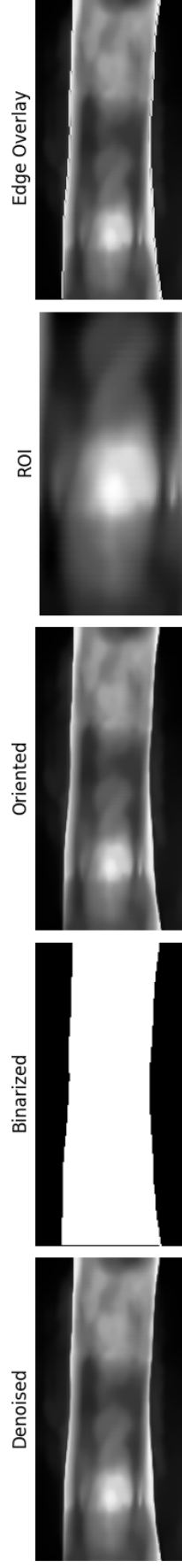


Figure 4.8: ROI Extracted and Preprocessed Image from ID 3, Vein 1, and Sample 1 in UTFVP Dataset

4.2 Augmentation of ROIs in UTFVP Dataset

During the preparation of the UTFVP dataset, the extracted 60×128 -pixel regions of interest (ROIs) were not kept in their original form. They were expanded through a simple augmentation routine meant to mimic the kinds of small disturbances that happen when finger-vein images are collected in practice. In routine acquisition, the lighting rarely stays constant, the finger may shift slightly, and the sensor can introduce its own noise. To reflect these effects without altering the underlying vein pattern, each ROI is processed three separate times using different randomly chosen transformations. The goal is not to create new structures but to give the model a broader set of slightly varied samples so it learns to focus on the stable parts of the pattern.

The augmentation pipeline $\mathcal{A}$ can be formally defined as a composite function applied to the original image I_o to produce an augmented image I_a , as shown in Equation 4.12:

$$I_a = \mathcal{A}(I_o) = f_n(\dots f_2(f_1(I_o, \theta_1), \theta_2), \dots, \theta_n) \quad (4.12)$$

where f_i represents a specific augmentation function and θ_i its corresponding randomly sampled parameters.

4.2.1 Luminance Transformation

To simulate variations in illumination intensity during capture, a random brightness adjustment is applied in the HSV color space. The grayscale image is first converted to a three-channel BGR format and then to HSV. The Value (V) channel, which represents lightness, is scaled by a random factor.

The transformation is defined by Equations 4.13 and 4.14:

$$\alpha \sim \mathcal{U}(0.6, 1.5) \quad (4.13)$$

$$V'(x, y) = \min(\max(\alpha \cdot V(x, y), 0), 255) \quad (4.14)$$

where α is a factor sampled from a uniform distribution $\mathcal{U}$ between 0.6 and 1.5, and $V'(x, y)$ is the transformed luminance value at pixel (x, y) . The resulting image is then converted back to grayscale.

4.2.2 Random Rotation

Slight rotational misalignment of the finger is simulated by applying a random rotation around the image center. This ensures the model does not overfit to a perfectly upright finger orientation.

The rotation matrix M_r is defined by Equation 4.15:

$$M_r = R(\theta), \quad \text{where } \theta \sim \mathcal{U}(-10^\circ, 10^\circ) \quad (4.15)$$

where R is the standard 2D rotation matrix. The image is warped using M_r with reflective border handling.

4.2.3 Random Affine Translation

Minor translational shifts in the finger's position within the ROI are simulated using an affine transformation. This teaches the model to be invariant to small registration errors.

The translation is represented by the transformation matrix M_t in Equation 4.16:

$$M_t = \begin{bmatrix} 1 & 0 & \Delta x \\ 0 & 1 & \Delta y \end{bmatrix}, \quad \text{where } \Delta x, \Delta y \sim \mathcal{U}(-4, 4) \quad (4.16)$$

The transformed image I_t is then computed using Equation 4.17:

$$I_t = W(I, M_t) \quad (4.17)$$

where W denotes the affine warping function with border reflection to handle out-of-

frame pixels, as introduced by Zhao et al. [40].

4.2.4 Gaussian Noise Injection

Sensor noise and quantum effects are simulated by adding pixel-wise Gaussian noise.

The standard deviation of the noise is randomly varied for each augmentation to simulate different noise levels across capturing devices or conditions.

The noise injection is formalized in Equation 4.18:

$$I_{\text{noisy}}(x, y) = \min(\max(I(x, y) + \eta(x, y), 0), 255), \eta(x, y) \sim \mathcal{N}(0, \sigma^2) \quad (4.18)$$

$$\sigma \sim \mathcal{U}(10, 30) \quad (4.19)$$

Pixel $I(x, y)$ is perturbed by adding Gaussian noise η from $\mathcal{N}(0, \sigma^2)$, which has a mean of 0 and intensity σ . The result is clamped to 0-255 to produce the final noisy pixel $I_{\text{noisy}}(x, y)$.

4.2.5 Random Gaussian Blur

To simulate slight defocus or motion blur, a Gaussian smoothing filter with a randomly chosen kernel size is applied. This encourages the model to learn features that are robust to minor image softening.

The blur operation I_{blurred} is defined by Equation 4.20:

$$I_{\text{blurred}} = G_k * I, \quad \text{where } k \sim \mathcal{U}\{3, 5\} \quad (4.20)$$

Here, G_k is a 2D Gaussian kernel of size $k \times k$ with standard deviation derived from the kernel size, and $*$ denotes the convolution operation.

4.2.6 Random Occlusion

Real-world scenarios may involve partial obstructions (e.g., dirt, water droplets). This is simulated by adding a small black rectangle at a random position within the ROI, forcing the model to rely on features from multiple regions.

The occlusion operation I_{occ} is defined by Equation 4.21:

$$I_{\text{occ}}(y : y + h_o, x : x + w_o) = 0, \quad \text{where} \quad x \sim \mathcal{U}(0, W - w_o), y \sim \mathcal{U}(0, H - h_o) \quad (4.21)$$

where $h_o = 10$ and $w_o = 10$ are the fixed height and width of the occlusion patch, and H, W are the dimensions of the ROI.

4.2.7 Augmentation Pipeline and Output

The complete stochastic pipeline $\mathcal{A}$ for generating a single augmented image is the sequential application of all aforementioned functions, as summarized in Figure 4.9.

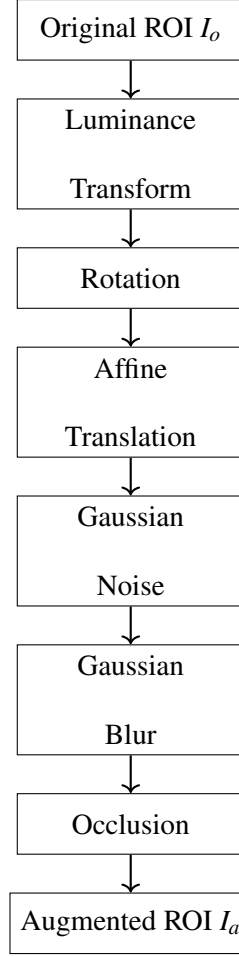


Figure 4.9: The Stochastic Augmentation Pipeline Applied Sequentially to Each Original ROI. Parameters for Each Step Are Randomly Sampled per Augmentation.

4.3 Vein Image Preprocessing Across FV-USM, MNCBNU-6000, and UTFVP Datasets

During the application of the defined protocols on the dataset, all extracted Regions of Interest (ROIs) undergo a consistent three-step preprocessing pipeline prior to feature extraction and classification. In Strategy 1 (multi-fusion), this preprocessing is applied to each individual ROI before fusing six fingers into a composite image, whereas in Strategy 2 (single-finger), It is applied independently to each ROI, without fusion. This design preserves finger-specific vein characteristics and avoids potential information dilution or bias that may arise from fusion-based approaches. Consequently, the recognition performance directly reflects the discriminative power of the features extracted from an individual finger. The sequential preprocessing steps are as follows:

1. Fast Non-Local Means Denoising

To suppress noise while preserving edges and fine texture details, we apply the Fast Non-Local Means denoising algorithm. The denoised value at pixel i is computed as Equation 4.22:

$$\hat{u}(i) = \sum_{j \in \mathcal{J}} w(i, j) v(j), \quad (4.22)$$

where the weights can be taken from Equation 4.23:

$$w(i, j) = \frac{1}{Z(i)} \exp \left(-\frac{\|v(N_i) - v(N_j)\|^2}{h^2} \right), \quad (4.23)$$

with normalization factor provided in 4.24:

$$Z(i) = \sum_{j \in \mathcal{J}} w(i, j). \quad (4.24)$$

Here, $v(j)$ is the observed intensity at pixel j , N_i and N_j denote local neighborhoods, $\mathcal{J}$ is the image domain, and h is the filtering parameter ($h = 10$ in this work).

2. Histogram Equalization

To improve global contrast, histogram equalization is applied. For an image with L discrete intensity levels, the transformation is shown in Equation 4.25:

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j), \quad (4.25)$$

where r_k is the k -th input intensity level, s_k is the corresponding output level, and $p_r(r_j)$ is the normalized histogram value. In our implementation, histogram equalization maps intensities to the range $[0, 1]$.

3. Standardization (Z-score Normalization)

Finally, each pixel value is standardized to have zero mean and unit variance, as given in Equation 4.26:

$$x' = \frac{x - \mu}{\sigma + \varepsilon}, \quad (4.26)$$

where x is the original intensity value of a given pixel, x' is the standardized intensity value, μ is the mean intensity value of all pixels in the image, and σ is

the standard deviation of the pixel intensities, defined as Equation 4.27.

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i, \quad \sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}, \quad (4.27)$$

with x_i denoting the intensity of the i -th pixel, N the total number of pixels in the image ($N = H \times W$), and $\varepsilon = 10^{-8}$ a small constant added to prevent division by zero.

4.4 Feature Extraction Methodologies

In this research, three distinct categories of feature extraction techniques were implemented in all of the strategies and protocols:

- **Dimension reduction methods :** including Conventional PCA, 2DPCA, and 2D²PCA.
- **Texture-based methods:** including Local Binary Patterns (LBP) with different parameter settings such as LBP_{1,8}, LBP_{2,8}, and LBP_{1,16}, as well as Multi-Radius Fused LBP.
- **A deep learning-based method:** AlexNet architecture.

Each method's explanation is provided in the following subsections.

4.4.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is used to map the high-dimensional feature vectors into a smaller space while keeping the dominant variations intact. In this work, PCA is applied to the finger-vein feature maps obtained after the preprocessing stage and, when required, after the fusion of multiple descriptors. Let the feature matrix be $\mathbf{X} \in \mathbb{R}^{n \times d}$, where each sample contributes one row and each feature occupies a column. Before applying PCA, we perform mean-centering across each feature dimension, resulting in the centered feature matrix $\mathbf{X}_c$, as shown in Equation 4.28:

$$\mathbf{X}_c = \mathbf{X} - \mathbf{1}_n \mu^\top \in \mathbb{R}^{n \times d} \quad (4.28)$$

In which μ is the mean of the d features and $\mathbf{1}_n$ is an $n \times 1$ vector of ones. Centering

ensures that the subsequent spectral decomposition reflects variance around the mean rather than absolute magnitudes, which is essential for the Gram-based formulation used later.

The PCA implementation consists of four key computational steps:

1. Gram Matrix Computation (Eq. 4.29)

$$\mathbf{G} = \mathbf{X}_c \mathbf{X}_c^\top \in \mathbb{R}^{n \times n} \quad (4.29)$$

In this formulation, the Gram matrix acts as a practical alternative to the full $d \times d$ covariance matrix. Instead of working directly in the original feature space, the Gram matrix summarizes how similar the samples are to one another. When $d \gg n$, which is the case for the fused finger-vein descriptors used here, the $n \times n$ matrix is far smaller than the covariance matrix and therefore much easier to store and process. This reduces the memory requirement from $O(d^2)$ to $O(n^2)$, while still preserving the eigen-information needed to compute the principal components. In this context, $\mathbf{X}_c$ is the centered feature matrix whose rows correspond to the fused ROI descriptors, and whose columns represent the d feature dimensions extracted from each image.

2. Eigenvalue Decomposition (Eq. 4.30)

$$\mathbf{G} \mathbf{u}_i = \lambda_i \mathbf{u}_i \quad (4.30)$$

At this point, we decompose the Gram matrix into its eigenvalues and eigenvectors. Because $\mathbf{G}$ is symmetric and positive semi-definite, this step behaves nicely in practice. Each eigenvalue λ_i tells us how much variation is carried by the i -th direction, and the associated eigenvector $\mathbf{u}_i$ gives that direction in the n -dimensional sample space. Once we sort the eigenvalues from largest to smallest, we immediately know which components matter most for representing the data.

3. Component Reconstruction (Eq. 4.31)

$$\mathbf{w}_i = \frac{\mathbf{X}_c^\top \mathbf{u}_i}{\sqrt{\lambda_i}} \in \mathbb{R}^d \quad (4.31)$$

After finding $\mathbf{u}_i$ in the sample space, we bring it back into the original feature space. This gives the principal components $\mathbf{w}_i$ in $\mathbb{R}^d$, meaning each $\mathbf{w}_i$ becomes one orthonormal axis of the PCA subspace. To keep things stable numerically, we only retain components with $\lambda_i > 10^{-10}$. In other words, we drop directions that are basically noise, while keeping every component that contributes real variance to the representation.

4. Dimensionality Reduction(Eq. 4.32)

$$\mathbf{Y} = \mathbf{X}_c \mathbf{W} \in \mathbb{R}^{n \times k} \quad (4.32)$$

The final projection matrix $\mathbf{W} = [\mathbf{w}_1 \cdots \mathbf{w}_k]$ contains the top k principal components selected to preserve 95% of the total cumulative variance.¹ The reduced-dimension representation $\mathbf{Y}$ maintains the essential discriminative structure of the original high-dimensional finger-vein features while enabling efficient computation for subsequent matching operations.

4.4.2 Two-Dimensional Principal Component Analysis (2DPCA)

Two-Dimensional Principal Component Analysis (2DPCA) is employed to reduce the dimensionality of finger-vein images while keeping both the dominant variance and their spatial arrangement. Unlike standard PCA, which vectorizes images into long 1-D feature vectors, 2DPCA operates directly on the image matrices, so the row-column structure is preserved throughout the projection.

Let the training set include n images $\{A_1, A_2, \dots, A_n\}$, where each $A_i \in \mathbb{R}^{H \times W}$ corresponds either to a fused multi-finger composite or to an individual finger image. Before applying 2DPCA, the images are mean-centered to remove sample bias, giving the centered matrices in Equation 4.33:

¹ Where λ_i are eigenvalues (descending order) of $\mathbf{X}_c^\top \mathbf{X}_c$, d =original dimension ($H \times W$), and k is the smallest integer satisfying $\frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^d \lambda_j} \geq 0.95$.

$$\mathbf{A}_c^{(i)} = \mathbf{A}_i - \bar{\mathbf{A}} \in \mathbb{R}^{H \times W} \quad (4.33)$$

where $\bar{\mathbf{A}} = \frac{1}{n} \sum_{i=1}^n \mathbf{A}_i$ is the mean image and $\mathbf{A}_c^{(i)}$ is the resulting mean-centered image.

This ensures zero-mean columns, a requirement for the covariance computation.

The 2DPCA implementation consists of four key computational steps:

1. Mean Image Computation (Eq. 4.34)

$$\bar{\mathbf{A}} = \frac{1}{n} \sum_{i=1}^n \mathbf{A}_i \in \mathbb{R}^{H \times W} \quad (4.34)$$

The mean image $\bar{\mathbf{A}}$ captures the average finger-vein pattern across all training samples. Each image is then centered by subtracting this mean pattern, maintaining the 2D structure of the data.

2. Image Covariance Matrix (Eq. 4.35)

$$\mathbf{G}_t = \frac{1}{n} \sum_{i=1}^n (\mathbf{A}_i - \bar{\mathbf{A}})^\top (\mathbf{A}_i - \bar{\mathbf{A}}) \in \mathbb{R}^{W \times W} \quad (4.35)$$

Unlike conventional PCA, 2DPCA computes the covariance matrix $\mathbf{G}_t$ along the image columns (width dimension), resulting in a compact $W \times W$ matrix. This formulation preserves row-wise (height) relationships while capturing column-wise variations.

3. Eigenvalue Decomposition (Eq. 4.36)

$$\mathbf{G}_t \mathbf{w}_j = \lambda_j \mathbf{w}_j \quad (4.36)$$

The symmetric positive semi-definite image covariance matrix $\mathbf{G}_t$ yields eigenvectors $\mathbf{w}_j$ (principal directions) sorted by descending eigenvalues λ_j . We retain the top- k components based on the criterion $\sum_{j=1}^k \lambda_j / \sum \lambda_j \geq 0.95$, which automatically determines the optimal reduced dimension.²

² The covariance matrix is defined as $\mathbf{G}_t = \frac{1}{n} \sum_{i=1}^n (\mathbf{A}_c^{(i)})^\top \mathbf{A}_c^{(i)}$. The projection reduces the image width from W to k while preserving the original height H , resulting in features $\mathbf{Y}_i \in \mathbb{R}^{H \times k}$.

4. Feature Projection (Eq. 4.37)

$$\mathbf{Y}_i = \mathbf{A}_i \mathbf{P} \in \mathbb{R}^{H \times k} \quad (4.37)$$

where $\mathbf{P} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_k] \in \mathbb{R}^{W \times k}$ is the projection matrix composed of the retained eigenvectors. The resulting feature matrix $\mathbf{Y}_i$ maintains the original image height H while reducing its width from W to k components. These matrices are subsequently flattened into vectors for classifier input, preserving spatial relationships more effectively than vectorized PCA.

4.4.3 Bidirectional Two-Dimensional Principal Component Analysis ((2D)²PCA)

Bidirectional Two-Dimensional PCA ((2D)²PCA) extends 2DPCA by simultaneously reducing dimensions along both rows and columns while preserving spatial relationships. This method is particularly effective for finger-vein recognition as it maintains the natural 2D structure of vein patterns while achieving greater dimensionality reduction.

Let the training set consist of n images $\{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_n\}$ where each $\mathbf{A}_i \in \mathbb{R}^{H \times W}$ represents either a fused multi-finger composite or individual finger image. The mean image $\bar{\mathbf{A}}$ is computed as Equation 4.38:

$$\bar{\mathbf{A}} = \frac{1}{n} \sum_{i=1}^n \mathbf{A}_i \in \mathbb{R}^{H \times W} \quad (4.38)$$

The (2D)²PCA implementation consists of five key computational steps:

1. Row and Column Covariance Matrices (Eqs. 4.39–4.40)

$$\mathbf{G}_r = \frac{1}{n} \sum_{i=1}^n (\mathbf{A}_i - \bar{\mathbf{A}})(\mathbf{A}_i - \bar{\mathbf{A}})^\top \in \mathbb{R}^{H \times H} \quad (4.39)$$

$$\mathbf{G}_c = \frac{1}{n} \sum_{i=1}^n (\mathbf{A}_i - \bar{\mathbf{A}})^\top (\mathbf{A}_i - \bar{\mathbf{A}}) \in \mathbb{R}^{W \times W} \quad (4.40)$$

These matrices capture row-wise $\mathbf{G}_r$ and $\mathbf{G}_c$ column-wise variations respectively, preserving spatial relationships in both dimensions.

2. Bidirectional Eigenvalue Decomposition (Eqs. 4.41–4.42)

$$\mathbf{G}_r \mathbf{u}_i = \lambda_i^r \mathbf{u}_i \quad (4.41)$$

$$\mathbf{G}_c \mathbf{v}_j = \lambda_j^c \mathbf{v}_j \quad (4.42)$$

yielding eigenvectors $\mathbf{u}_i$ for the row projection and $\mathbf{v}_j$ for the column projection. These vectors are ordered according to their eigenvalues, λ_i^r and λ_j^c , arranged in descending magnitude, since each eigenvalue reflects how much variance is captured along its corresponding principal direction.

3. **Component Selection** We retain k_r row and k_c column components³ that preserve 95% variance in each dimension.

4. **Projection Matrices Construction (Eq. 4.43)**

$$\mathbf{U} = [\mathbf{u}_1 \cdots \mathbf{u}_{k_r}] \in \mathbb{R}^{H \times k_r}, \quad \mathbf{V} = [\mathbf{v}_1 \cdots \mathbf{v}_{k_c}] \in \mathbb{R}^{W \times k_c} \quad (4.43)$$

5. **Bidirectional Projection (Eq. 4.44)**

$$\mathbf{Y}_i = \mathbf{U}^\top \mathbf{A}_i \mathbf{V} \in \mathbb{R}^{k_r \times k_c} \quad (4.44)$$

The resulting features maintain maximal spatial variance while achieving compact representation.

4.4.4 Rotational-Invariant Uniform Local Binary Patterns (LBP^{riu2})

Local Binary Patterns with rotation-invariant uniform mapping (LBP^{riu2}) capture fine micro-texture features in finger-vein images by comparing the intensities of neighboring pixels. This study implements three configurations, namely LBP^{riu2}_(8,1), LBP^{riu2}_(16,1), and LBP^{riu2}_(8,2), each computed over the image using a sliding-window approach.

For each pixel (x_c, y_c) in image I , the basic LBP code is computed as Equation 4.45:

$$\text{LBP}_{P,R}(x_c, y_c) = \sum_{p=0}^{P-1} s(g_p - g_c) 2^p \quad (4.45)$$

³ Where k_r and k_c are determined by $\frac{\sum_{i=1}^{k_r} \lambda_i^r}{\sum \lambda^r} \geq 0.95$ and $\frac{\sum_{j=1}^{k_c} \lambda_j^c}{\sum \lambda^c} \geq 0.95$ respectively, with $k_r \leq H$ and $k_c \leq W$.

where g_c is the center pixel intensity, g_p are neighbor intensities sampled at radius R with P points, and $s(\cdot)$ is the sign function shown in Equation 4.46:

$$s(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (4.46)$$

The fundamental rotation invariance is achieved by identifying the canonical (minimum) representation of each binary pattern across all possible rotations. For a pattern i , its P -bit binary representation is rotated P times, and the lexicographically smallest rotation is selected as the canonical form. This ensures all rotations of the same fundamental pattern are mapped to the same code.

The RIU2 mapping then categorizes these canonical patterns by their uniformity (U), which counts the number of bitwise transitions in the circular sequence. The uniformity measure U is defined in Equation 4.47:

$$U(\text{LBP}) = |s(g_{P-1} - g_c) - s(g_0 - g_c)| + \sum_{p=1}^{P-1} |s(g_p - g_c) - s(g_{p-1} - g_c)| \quad (4.47)$$

Patterns with $U \leq 2$ are considered uniform. The final RIU2 value is assigned as shown in Equation 4.49:

$$\text{LBP}_{P,R}^{\text{RIU2}} = \begin{cases} \sum_{p=0}^{P-1} s(g_p - g_c) & \text{if } U(\text{LBP}) \leq 2 \\ P + 1 & \text{otherwise} \end{cases} \quad (4.48)$$

4.4.5 Multi-Radius Fused LBP^{riu2}

This study uses an extended version of the rotation-invariant LBP^{riu2} descriptor. Instead of computing features from only one neighbourhood setting, the descriptor is formed by fusing three complementary (P, R) configurations. The motivation is simple. Finger-vein ROIs contain meaningful texture cues at different spatial scales, and a single radius cannot capture all of them reliably. In the proposed multi-radius variant, the configuration with $(8, 1)$ responds primarily to high-frequency micro-texture, the configuration with $(16, 1)$ provides a denser and more detailed

angular sampling of the local pattern, and the configuration with (8,2) captures broader macro-structural information. These configurations are fused at the histogram level to construct the final multi-radius descriptor. The practical advantages of this design are summarized below:

- **Multi-Scale Representation:** By combining the three neighbourhood settings, the descriptor becomes sensitive to texture information across several spatial frequencies. The (8,1) configuration captures the finer vein patterns at roughly 50–100, μm , while (8,2) responds to broader structures around 200–400, μm . The (16,1) setting, with its denser angular sampling, helps the descriptor pick up branching behaviour and smaller directional changes that might otherwise be overlooked. When these responses are merged, the result is a representation that does not lean too heavily on either micro or macro details, and instead keeps a reasonable balance between the two.
- **Enhanced Feature Discriminability:** The fused histogram yields a compact 38-dimensional feature vector, where the three configurations contribute $10 + 18 + 10$ bins respectively. Even with this relatively low dimensionality, the combined representation separates classes more effectively than single-configuration baselines. Each histogram carries different pieces of the texture information, and merging them at the histogram level strengthens their discriminative power without increasing computational cost.
- **Preserved Invariance Properties:** All three configurations remain rotation-invariant because they enforce the uniformity constraint, $U(\text{LBP}) \leq 2$. The RIU2 mapping given in Equation 4.49 applies this same constraint at each scale, so the rotation invariance is preserved consistently across the multi-radius settings.

$$\text{LBP}_{P,R}^{\text{RIU2}} = \begin{cases} \sum_{p=0}^{P-1} s(g_p - g_c) & U(\text{LBP}) \leq 2 \\ P + 1 & \text{otherwise} \end{cases} \quad (4.49)$$

This means the descriptor keeps the same rotation-invariant behaviour regardless of which (P, R) setting is being used, while still benefiting from the added multi-

scale coverage.

The algorithm works patch by patch, using a 10×10 window with a stride of 10 pixels. For each patch, it builds three circular neighbourhoods at different scales: 8 points at $R = 1$ (about $7.5, \mu\text{m}$ spacing), 16 points again at $R = 1$ to get a denser angular sampling (around $3.75, \mu\text{m}$ resolution), and a third set of 8 points at $R = 2$ (roughly $15, \mu\text{m}$ spacing). These neighbourhood responses are then consolidated into a single patch descriptor by merging their histograms, following the fusion rule given in Eq. 4.50.

$$\mathbf{F}_{\text{patch}} = \mathbf{h}^{(8,1)} \oplus \mathbf{h}^{(16,1)} \oplus \mathbf{h}^{(8,2)} \in \mathbb{R}^{38} \quad (4.50)$$

where $\mathbf{h}^{(P,R)}$ represents the RIU2-LBP histogram for configuration (P,R) , with dimensionalities $\dim(\mathbf{h}^{(8,1)}) = 10$, $\dim(\mathbf{h}^{(16,1)}) = 18$, and $\dim(\mathbf{h}^{(8,2)}) = 10$. The $\oplus$ operator performs vector concatenation.

The window then moves across the ROI without overlap, so every valid patch is processed once. After extracting all patch descriptors, they are combined to form the final image-level feature according to 4.51:

$$\mathbf{F}_{\text{image}} = \bigcup_{i=1}^{N_p} \mathbf{F}_{\text{patch}}^i \quad (4.51)$$

where N_p represents the total number of patches taken from each ROI. This value depends directly on the ROI dimensions and on the geometry of the sliding patch grid. With this setup, the method captures texture information at several scales while still preserving the spatial layout of the underlying vascular structures.

4.4.6 Dimensionality Reduced Multi-Radius Fused LBP^{riu2}

The integration of multi-radius LBP features with dimensionality-reduction proceeds in a stepwise manner, aiming to retain the distinctive texture cues while reducing the size of the feature representation. The process starts with constructing the

multi-radius LBP descriptors. For every 10×10 patch in the image, rotation-invariant uniform LBP (LBP^{riu2}) histograms are extracted for three neighbourhood configurations: $(R = 1, P = 8)$, $(R = 1, P = 16)$, and $(R = 2, P = 8)$. The three histograms obtained from each patch are placed side by side, forming a single feature vector that captures information from both fine and coarse texture scales.

The patches are evaluated across the image using a stride of 10 pixels. As a result, each finger image is converted into a feature matrix whose rows correspond to specific patch positions, while the columns store the concatenated histogram bins. This arrangement preserves the spatial layout of the patterns—an aspect that later helps the PCA-based methods operate more effectively.

Our study considers two organisation strategies:

- **Strategy 1:** All six fingers (left/right forefinger, middle, and ring) are combined by concatenating their feature matrices horizontally to form a single composite matrix.
- **Strategy 2:** Each finger is processed independently with no cross-finger concatenation.

Both strategies keep the patch-level structure intact: rows continue to indicate spatial locations and columns maintain the sequence of histogram bins. Strategy 1 produces a larger composite representation capable of capturing cross-finger venous patterns, whereas Strategy 2 focuses strictly on finger-specific characteristics. Regardless of the strategy used, the resulting matrices are subsequently fed into the three PCA-based dimensionality-reduction variants.

Accordingly, this pipeline yields three dimensionality-reduced variants:

$$\begin{aligned} &\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}} + \text{PCA}, & \text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}} + 2\text{DPCA}, & \text{and} \\ &\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}} + 2\text{D}^2\text{PCA}. \end{aligned}$$

4.4.7 AlexNet

AlexNet-based Convolutional Neural Networks have proven particularly effective as feature extractors for finger vein biometric recognition. In this study, the classic AlexNet architecture, as shown in Figure 4.10, is adapted to process single-channel vein pattern images while being implemented in the two strategies. This adaptation maintains the powerful feature learning capabilities of network while optimizing it for the specific requirements of vein pattern analysis.

The modified network architecture is designed to preserve the spatial structure of the input by adjusting the dimensions of the convolutional and pooling layers with care. Although both strategies rely on the same underlying network weights, they differ in the way the input images are prepared and in how the final classification layers are arranged. These differences reflect the distinct assumptions each strategy makes about how vein-pattern information should be represented before entering the classifier.

4.4.7.1 Network Architecture

The model in this study is a tailored version of AlexNet, redesigned for grayscale finger-vein recognition. As in the original architecture, it is organized into two main parts: a feature extractor and a classifier. While the overall AlexNet backbone is preserved, we introduced targeted adjustments so the network can work effectively with single-channel vein images and, more importantly, learn robust and consistent representations that support reliable identification.

- **Convolutional Feature Extraction:** The feature extractor consists of five convolutional layers. Each layer applies the standard convolution operation shown in Equation 4.52:

$$\mathbf{Y}^{(l)}(x,y) = \mathbf{B}^{(l)} + \sum_{c=1}^{C_{in}} \sum_{i=-\kappa}^{\kappa} \sum_{j=-\kappa}^{\kappa} \mathbf{W}^{(l)}(c,i,j) * \mathbf{X}^{(l-1)}(x+i,y+j) \quad (4.52)$$

In this formulation, $\mathbf{Y}^{(l)}(x,y)$ denotes the output produced at spatial location (x,y) in layer l . The learnable convolutional kernels are given by $\mathbf{W}^{(l)}$, while $\mathbf{B}^{(l)}$ acts as the corresponding bias term, and $\mathbf{X}^{(l-1)}$ refers to the input feature maps

coming from the previous layer. Because the network operates on grayscale images, $C_{\text{in}} = 1$ in the first layer, and κ specifies the kernel radius (i.e., the size of the local neighborhood used in the convolution).

- **Batch Normalization:** To support more stable training and help the network converge faster, batch normalization is applied after each convolutional layer. Its operation follows the formulation in Equation 4.53:

$$\hat{\mathbf{X}} = \frac{\mathbf{X} - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \varepsilon}} \quad ; \quad \mathbf{Y} = \gamma \hat{\mathbf{X}} + \beta \quad (4.53)$$

where $\mu_{\mathcal{B}}$ and $\sigma_{\mathcal{B}}^2$ are the batch mean and variance, $\varepsilon = 10^{-5}$ ensures numerical stability, and γ, β are learnable parameters initialized to 1 and 0.

- **Spatial Downsampling:** Max pooling is applied to gradually reduce the spatial resolution of the feature maps while preserving the strongest and most informative responses. The pooling operation follows the formulation in Equation 4.54.

$$\mathbf{P}(x, y) = \max_{i, j \in \mathcal{R}} \mathbf{X}(x \cdot s + i, y \cdot s + j) \quad (4.54)$$

The output value $\mathbf{P}(x, y)$ is taken from a local region $\mathcal{R}$, and the stride s determines the downsampling rate.

- **Classifier Design:** The classifier is built from two fully connected layers, and dropout is used during training to regularize the model and reduce overfitting. The dropout operation is defined in Equation 4.55.

$$\mathbf{y}_{\text{train}} = \mathbf{z} \odot \mathbf{m} \quad \text{where} \quad m_i \sim \text{Bernoulli}(1 - 0.5) \quad (4.55)$$

The vector $\mathbf{m}$ contains binary samples drawn independently with probability $1 - p$, where $p = 0.5$ in our experiments. By randomly masking activations in this way, the classifier is discouraged from depending too strongly on any single unit.

- **Weight Initialization:** Proper initialization is critical for training stability. Equations 4.56 and 4.57 are showing He (or Kaiming) and Xavier (or Glorot) ,

respectively:

$$\text{Convolutional: } \mathbf{W}_{\text{conv}} \sim \mathcal{N}\left(0, \sqrt{\frac{2}{C_{\text{in}} \times \kappa^2}}\right) \quad (4.56)$$

$$\text{Fully-Connected: } \mathbf{W}_{\text{fc}} \sim \mathcal{U}\left(-\sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}, \sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}\right) \quad (4.57)$$

For Convolutional layers, C_{in} is input channels, κ the kernel size, and $\mathbf{W}_{\text{conv}}$ has shape $(C_{\text{out}}, C_{\text{in}}, \kappa, \kappa)$. For Fully-connected layers, n_{in} (e.g., $256 \times 6 \times 6 = 9216$) and n_{out} (e.g., 4096) define $\mathbf{W}_{\text{fc}}$'s $(n_{\text{out}}, n_{\text{in}})$ matrix. $\mathcal{N}(\mu, \sigma)$ and $\mathcal{U}(a, b)$ denote normal/uniform distributions respectively.

4.4.8 AlexNet Implementation for the Two Strategies

The architecture in this study can be run in two practical modes for finger-vein identification, chosen depending on how the system is deployed. In both cases, the same convolutional backbone is used for feature extraction; the difference is in how the inputs are organized and how the final classifier is defined. Concretely, we implement:

- **Strategy 1. Fused Finger Representation:** In this setting, all fingers of a subject are considered together to perform identity-level recognition. The input tensor is arranged as $[batch_size, num_fingers, 1, 224, 224]$. Each finger image is passed through the shared convolutional feature extractor, yielding $256 \times 6 \times 6$ feature maps per finger. After flattening each map into a 9216-dimensional vector, the vectors from all fingers are concatenated to form a single fused representation. This combined vector is then fed into the fully connected classifier, which predicts the subject identity (with the number of output classes matching the number of subjects). The motivation here is simple: by seeing multiple fingers at once, the model can exploit complementary vein cues across them.

Strategy 2. Per-Finger Classification: In this strategy, each finger is handled as its own biometric sample, so the model performs recognition at the finger level rather than at the full-identity level. A single input consists of one finger image with dimensions $[batch_size, 1, 224, 224]$. The convolutional backbone used in

the fused approach is kept here as well, but now it processes each finger on its own. Once the final convolutional layer completes, the output is reshaped into a 9216-dimensional feature vector. This vector is passed to the fully connected classifier, which links it to one of the finger-specific categories. The total number of classes is determined by multiplying the number of subjects by the number of fingers. Both training and evaluation take place at this finger-based level, and the shared convolutional weights let the network pick up broader patterns that appear across the dataset.

4.4.8.1 Training Protocol

To keep the comparison across strategies consistent, the model is trained under the same configuration in every case. The learning objective is the categorical cross-entropy loss, which quantifies how far the predicted distribution $\hat{\mathbf{y}}$ is from the true label vector $\mathbf{y}$. Its standard form is shown in Eq. 4.58:

$$\mathcal{L}_{\text{CE}} = - \sum_{c=1}^C y_c \log \hat{y}_c \quad (4.58)$$

where, C is the number of classes, $y_c \in \{0, 1\}$ indicates whether class c is the correct one, and $\hat{y}_c$ is the model's softmax probability for that class. This loss encourages the model to place high probability on the correct class, and it increases the penalty whenever the predicted distribution drifts away from the ground truth.

For optimization, SGD was used with momentum. The momentum term is set to 0.9, and weight decay is fixed at 10^{-4} . In practice, this combination keeps the updates from becoming noisy and adds a small regularizing push that helps with generalization. Training is paired with a OneCycleLR learning-rate schedule: the learning rate ramps up to a peak of $\eta_{\text{max}} = 0.1$ and then is reduced gradually over the 50 epochs. The idea is to let the model move quickly through the parameter space at the start, and then tighten the steps later for fine adjustment. The update rule itself follows Eq. 4.59; briefly, the velocity v_t accumulates recent gradient information, the weight decay term acts directly on the parameters, and the step size η_t is supplied by the scheduler.

$$v_t = \mu \cdot v_{t-1} + \nabla \mathcal{L}(\theta_t) + \lambda \cdot \theta_t, \quad \theta_{t+1} = \theta_t - \eta_t \cdot v_t, \quad (4.59)$$

where $\mu = 0.9$ is the momentum term, $\lambda = 10^{-4}$ controls the weight decay, v_t is the accumulated velocity, $\nabla \mathcal{L}(\theta_t)$ is the gradient at iteration t , and η_t is the learning rate supplied by the scheduler.

The OneCycleLR schedule spans all T iterations. The first segment, $T_{\text{up}} = \lfloor 0.3T \rfloor$, increases the learning rate gradually, while the remaining iterations form the annealing phase. During these stages, the learning rate moves between $\eta_{\text{min}} = 0.001$ and a final value of 10^{-5} , enabling faster progress in the early stages of optimization and more controlled convergence near the end of training.

To encourage better generalization, dropout with probability $p = 0.5$ is applied in the fully connected layers. Batch normalization is also used throughout the network to keep the activations well-scaled and to reduce the internal covariate shift that can otherwise slow down learning. All experiments rely on preprocessed Region-of-Interest (ROI) images so that comparisons with dimensionality-reduction approaches and texture-based methods are conducted under identical and controlled conditions.

4.4.8.2 Evaluation & Deployment

During evaluation, the network is switched to evaluation mode, meaning gradients are disabled and no parameters are updated. Under Strategy 1, performance is reported as subject-level identification accuracy based on the fused multi-finger representation. Under Strategy 2, we instead assess recognition at the finger level and report per-finger classification accuracy. From an implementation standpoint, a few details are important for stable and fair testing: (i) variable image resolutions are handled using adaptive pooling with a 3×3 kernel and stride of 2, so the feature dimensions remain consistent; (ii) class imbalance is mitigated through weighted sampling during evaluation; and (iii) data augmentation is applied carefully to preserve the anatomical

continuity of vein patterns. A summary of these settings is provided in Table 4.1.

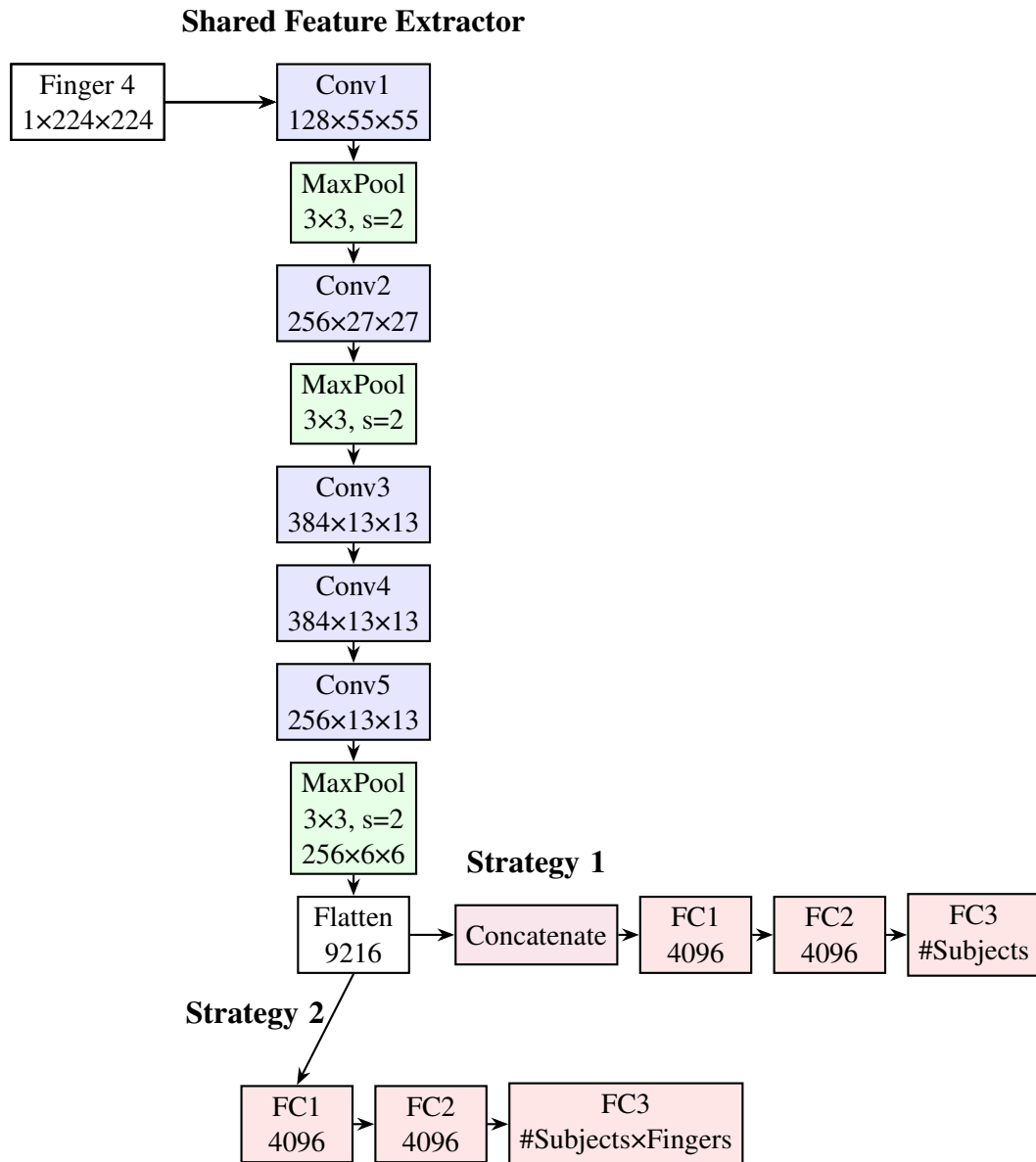


Figure 4.10: Adapted AlexNet Architecture for Finger Vein Identification, Illustrating the Two Operational Strategies: Strategy 1 (Fused Finger Representation) and Strategy 2 (Per-Finger Classification). The Shared Feature Extractor Processes Inputs From Multiple Fingers, With Strategy-Dependent Classifier Paths.

Table 4.1: Comparison of AlexNet Identification Strategies

Characteristic	Strategy 1	Strategy 2
Input Tensor Dimensions	Batch size $\times$ Number of fingers $\times$ 1 channel $\times$ 224 pixels $\times$ 224 pixels	Batch size $\times$ 1 channel $\times$ 224 pixels $\times$ 224 pixels
Label Encoding	Label = Subject ID - 1	Label = (Subject ID - 1) $\times$ Number of fingers + (Finger index - 1)
Classification Target	Subject identity	Individual finger identity
Number of Output Classes	Total number of subjects	Total number of subjects $\times$ Number of fingers per subject
Feature Processing	Simultaneous processing of all fingers with feature concatenation	Independent processing of single fingers

Chapter 5

IDENTIFICATION RESULTS

This chapter reports and interprets the identification performance achieved by the proposed finger vein recognition pipeline on the University of Twente Finger Vein Pattern (UTFVP), Finger Vein University of Sains Malaysia (FV-USM), and Multi-Modal Contactless Biometric Database Number 6000 (MMCBNU-6000) datasets, using the two evaluation strategies and three protocols defined earlier. All samples are processed through the same preprocessing routine introduced in Chapter 4, including region-of-interest (ROI) extraction, noise suppression, contrast enhancement, and final intensity standardization, so that the reported differences can be attributed primarily to the feature representation and matching stage rather than inconsistent input preparation.

The experimental design combines classical subspace learning methods and texture-based descriptors with a deep learning baseline: Principal Component Analysis (PCA) retains all non-zero eigenvectors in its Gram-matrix formulation, while Two-Dimensional Principal Component Analysis (2DPCA) and Bidirectional Two-Dimensional Principal Component Analysis (2D²PCA) are evaluated using a fixed dimensionality of 137 components to keep comparisons consistent across protocols and strategies. In parallel, Rotation-Invariant Uniform Local Binary Patterns (LBP^{riu2}) are tested in several neighborhood configurations, together with the proposed Multi-Radius Fused Rotation-Invariant Uniform Local Binary Patterns (Multi-Radius Fused LBP^{riu2}) and its dimensionality-reduced variants, and these hand-crafted methods are contrasted against an adapted AlexNet convolutional neural network trained and evaluated under the same protocol definitions.

Under Strategy 1, finger images are first fused and then processed by the corresponding feature extractor (including the proposed descriptor under the same fuse-then-extract workflow), whereas Strategy 2 operates directly on per-finger samples to assess finger-level discriminability. The resulting Rank-1 identification rates are summarized for all datasets and evaluation settings, followed by Cumulative Match Characteristic analysis for Strategy 2 and a comparative discussion against previously reported baselines and convolutional neural network architectures in the literature.

5.1 Results and Discussions

All images from the UTFVP, FV-USM, and MMCBNU-6000 datasets went through the preprocessing routine described in Chapter 4. This included ROI extraction, several denoising steps, contrast adjustments, and a final normalization stage. The identification experiments themselves followed the strategies and protocols outlined earlier in Chapter 3. For PCA, every non-zero eigenvector was kept, whereas 2DPCA and $(2D)^2PCA$ relied on a fixed set of 137 components so that the comparisons remained fair. These settings were not chosen arbitrarily; they were tuned empirically after multiple trial runs. Under Strategy 1, the finger images were first fused and then fed into a classical feature extractor, and the proposed method followed the same fuse-then-extract workflow. The Rank-1 identification results for each experiment appear in Tables 5.1, 5.2, 5.3, and 5.4.

Table 5.1: Session-Sensitive Evaluation for FV-USM

Method	Strategy 1			Strategy 2		
	Protocol 1	Protocol 2	Protocol 3	Protocol 1	Protocol 2	Protocol 3
PCA	89.02	86.04	91.06	79.88	77.68	83.54
2DPCA	89.02	85.77	91.46	80.89	78.08	83.64
2D ² PCA	90.45	86.99	90.65	76.37	73.14	80.08
LBP ^{riu2} _{(8,1),(16,1),(8,2)}	96.95	94.04	97.56	90.45	87.80	92.68
LBP ^{riu2} _(8,1)	97.36	94.72	97.56	91.06	88.48	93.80
LBP ^{riu2} _(16,1)	97.15	94.85	97.56	90.35	87.20	92.38
LBP ^{riu2} _(8,2)	96.14	93.22	97.56	89.08	86.89	92.07
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + PCA	94.72	92.55	95.12	85.16	81.64	89.13
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + 2DPCA	95.73	93.09	96.75	88.92	85.74	91.57
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + 2D ² PCA	91.46	91.33	89.84	89.43	86.79	91.87
AlexNet	84.15	79.27	87.40	79.17	75.61	82.48

Table 5.2: Session-Independent Evaluation for FV-USM

Method	Strategy 1			Strategy 2		
	Protocol 1	Protocol 2	Protocol 3	Protocol 1	Protocol 2	Protocol 3
PCA	100.00	100.00	100.00	97.76	97.73	98.48
2DPCA	100.00	100.00	100.00	97.61	97.39	98.78
2D ² PCA	100.00	100.00	100.00	94.56	93.26	95.93
LBP ^{riu2} _{(8,1),(16,1),(8,2)}	100.00	100.00	100.00	98.73	98.41	99.49
LBP ^{riu2} _(8,1)	100.00	100.00	100.00	98.73	98.20	99.49
LBP ^{riu2} _(16,1)	100.00	100.00	100.00	98.42	97.73	99.39
LBP ^{riu2} _(8,2)	100.00	100.00	100.00	98.93	98.54	99.49
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + PCA	100.00	100.00	100.00	97.76	96.68	98.78
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + 2DPCA	100.00	100.00	100.00	98.73	97.95	99.19
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + 2D ² PCA	100.00	100.00	100.00	99.19	98.68	99.59
AlexNet	100.00	99.73	100.00	99.14	98.71	99.80

Table 5.3: One Session Evaluation in UTFVP

Method	Strategy 1			Strategy 2		
	Protocol 1	Protocol 2	Protocol 3	Protocol 1	Protocol 2	Protocol 3
PCA	100.00	100.00	100.00	59.94	52.53	73.61
2DPCA	100.00	100.00	100.00	94.39	89.03	96.81
2D ² PCA	100.00	99.79	100.00	66.17	55.59	79.44
LBP ^{riu2} _{(8,1),(16,1),(8,2)}	100.00	100.00	100.00	94.39	88.06	98.47
LBP ^{riu2} _(8,1)	100.00	100.00	100.00	91.39	82.95	95.83
LBP ^{riu2} _(16,1)	100.00	100.00	100.00	93.83	85.90	96.81
LBP ^{riu2} _(8,2)	100.00	100.00	100.00	93.22	85.80	98.06
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + PCA	100.00	100.00	100.00	89.89	79.41	95.28
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + 2DPCA	100.00	100.00	100.00	92.17	84.13	97.36
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + 2D ² PCA	99.67	99.79	100.00	90.28	79.86	96.53
AlexNet	100.00	100.00	100.00	99.06	98.12	100.00

Table 5.4: One Session Evaluation in MMCBNU_6000

Method	Strategy 1			Strategy 2		
	Protocol 1	Protocol 2	Protocol 3	Protocol 1	Protocol 2	Protocol 3
PCA	100.00	100.00	100.00	98.50	96.40	99.33
2DPCA	100.00	100.00	100.00	98.44	96.67	99.17
2D ² PCA	100.00	100.00	100.00	98.61	96.57	99.33
LBP ^{riu2} _{(8,1),(16,1),(8,2)}	100.00	100.00	100.00	99.39	98.47	99.50
LBP ^{riu2} _(8,1)	100.00	100.00	100.00	99.33	98.57	99.50
LBP ^{riu2} _(16,1)	100.00	100.00	100.00	99.28	98.60	99.50
LBP ^{riu2} _(8,2)	100.00	100.00	100.00	97.83	95.33	99.50
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + PCA	100.00	100.00	100.00	98.78	97.07	99.00
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + 2DPCA	100.00	100.00	100.00	99.22	98.33	99.50
LBP ^{riu2} _{(8,1),(16,1),(8,2)} + 2D ² PCA	100.00	100.00	100.00	99.50	98.83	99.50
AlexNet	100.00	100.00	100.00	100.00	100.00	100.00

The results show a clear data-dependent pattern in descriptor effectiveness. In scarce-data, session-sensitive settings such as FV-USM, hand-crafted texture descriptors substantially outperform the AlexNet CNN baseline. Under Strategy 1, LBP variants are the best performers. Combining LBP descriptors with dimensionality-reduction methods reduces accuracy by about 2.82% on average across all protocols. Under the same strategy, AlexNet performs 12.61 percentage points worse than the LBP variants on average. Among the LBP variants, the multi-radius fused descriptor $\text{LBP}_{(8,1)(16,1)(8,2)}^{\text{riu2}}$ underperforms the single-radius variants $\text{LBP}_{(8,1)}^{\text{riu2}}$ and $\text{LBP}_{(16,1)}^{\text{riu2}}$ by approximately 0.34% on average across all protocols. Under Strategy 2 (all protocols), LBP variants achieve between 86.06% and 93.80%, outperforming other feature extractors, including LBP variants that have been combined with dimensionality-reduction methods. In Strategy 2, AlexNet is on average 11.09 percentage points lower than the LBP variants.

Dimensionality-reduction consistently degrades the discriminability of fused LBP in scarce-data settings. For Strategy 1, Protocol 1, raw multi-radius $\text{LBP}_{(8,1)(16,1)(8,2)}^{\text{riu2}}$ attains 96.95%, while multi-radius fused LBP + PCA yields 94.72% ($\Delta = 2.23\%$) and multi-radius fused LBP + 2DPCA attains 95.73% ($\Delta = 1.22\%$). multi-radius fused LBP + 2D²PCA produces less consistent results (91.46%). Thus, the $\text{LBP}_{(8,1)(16,1)(8,2)}^{\text{riu2}}$ + PCA variants and AlexNet tend to underperform when training data are limited. The same degradation pattern appears when the protocol is fixed but the strategy differs: raw multi-radius fused $\text{LBP}_{(8,1)(16,1)(8,2)}^{\text{riu2}}$ achieves 90.45%, whereas fused LBP + PCA drops to 85.16% ($\Delta = 5.29\%$) and fused LBP + 2DPCA reaches 88.92% ($\Delta = 1.53\%$); fused LBP + 2D²PCA yields 89.43% ($\Delta = 1.02\%$).

Conversely, when sufficient within-subject training samples are available—such as in the UTFVP and MMCBNU-6000 one-session protocols or in FV-USM session-independent evaluations—AlexNet achieves top or near-top performance. In Strategy 1, all methods in the FV-USM session-independent experiments approach

ceiling performance. The only exception is Protocol 2 of Strategy 1, where AlexNet reached 99.73%, largely due to the limited amount of training data. Under Strategy 2, PCA-based variants attain between 93.26% and 98.78%; however, LBP variants consistently outperform them across all protocols. On top of the LBP variants, AlexNet achieves between 99.14% and 99.80%, establishing itself as the most effective feature extractor in these settings. Among the LBP methods, $\text{LBP}_{(8,2)}^{\text{riu2}}$ surpasses the other variants, outperforming the proposed multi-radius fused descriptor $\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}}$ by approximately 0.20% in Protocol 1 and 0.13% in Protocol 2.

Notably, among the classical methods, our proposed multi-radius fused LBP combined with $2\text{D}^2\text{PCA}$ achieves superior performance across all protocols under Strategy 2. In Protocol 1 of Strategy 2, this method even outperformed the adapted AlexNet used in this study by 0.05%, although in Protocol 3 it underperforms by about 0.21%. In Strategy 1 of the UTFVP dataset, most methods achieve 100% accuracy, with the exception of $\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}} + 2\text{D}^2\text{PCA}$ in Protocols 1 and 2, and $2\text{D}^2\text{PCA}$ in Protocol 2. Interestingly, for $\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}} + 2\text{D}^2\text{PCA}$, Protocol 2 slightly outperforms Protocol 1 by 0.12%. Under Strategy 2, the proposed multi-radius fused descriptor $\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}}$ consistently outperforms all dimensionality-reduction techniques as well as the other implemented LBP variants. It also achieves higher accuracy compared to its own dimensionality-reduced versions. However, when compared with the adapted AlexNet, the superiority of the CNN-based method becomes evident. This adapted version of AlexNet achieves, on average, 5.42% higher accuracy across all protocols in Strategy 2.

In Strategy 1 of the MMCBNU-6000 dataset, all methods achieve 100% accuracy, which can be attributed to the compensation effect of finger samples being fused together. In Strategy 2, Protocol 3, most LBP variant methods attain 99.50%, outperforming the dimensionality-reduction-based methods. Notably, the adapted AlexNet achieves 100% accuracy across all protocols under the same strategy.

Among the classical feature extraction techniques, $\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu}2} + 2\text{D}^2\text{PCA}$ achieves 99.50% in Strategy 2, Protocol 1, outperforming the other classical methods.

Although single-radius variants (particularly $\text{LBP}_{(8,1)}^{\text{riu}2}$) achieve higher accuracy in certain session-sensitive cases such as FV-USM, the proposed multi-radius fused descriptor $\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu}2}$ consistently outperforms both LBP and PCA-based variants overall. When combined with dimensionality-reduction techniques, the proposed method can further enhance performance in specific scenarios, such as the one-session evaluation of MMCBNU-6000 and the session-independent protocols of FV-USM. Taken together, these results demonstrate that the proposed method delivers competitive performance across both scarce-data and data-abundant conditions.

5.2 Cumulative Match Characteristic Analysis for Strategy 2

The Cumulative Match Characteristic (CMC) curves provide critical insights into ranking performance beyond top-1 identification rates. Figures 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.7, 5.8, 5.9, 5.10, 5.11, and 5.12 presents CMC curves for all evaluated datasets under Strategy 2 protocols, revealing three key patterns consistent with our tabular results:

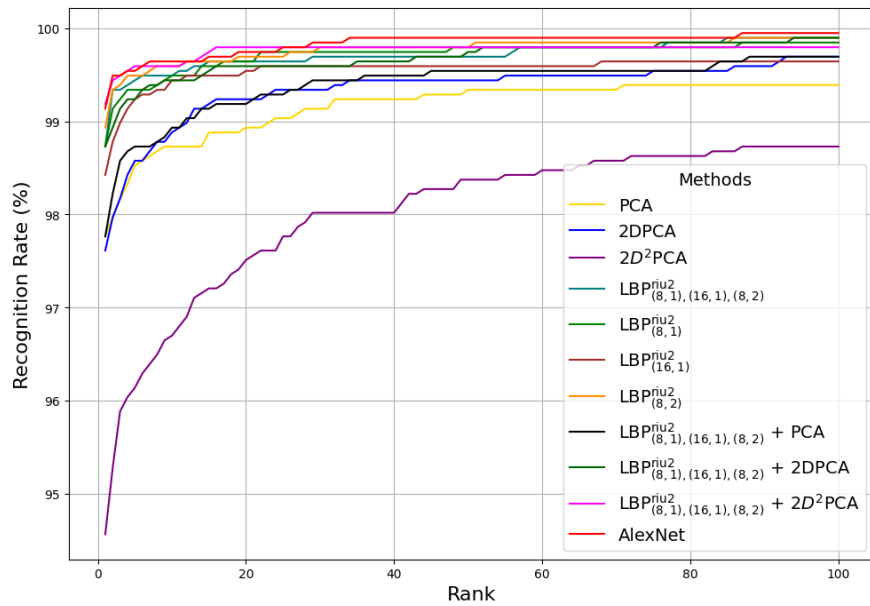


Figure 5.1: FVUSM Session Independent (S2_P1)

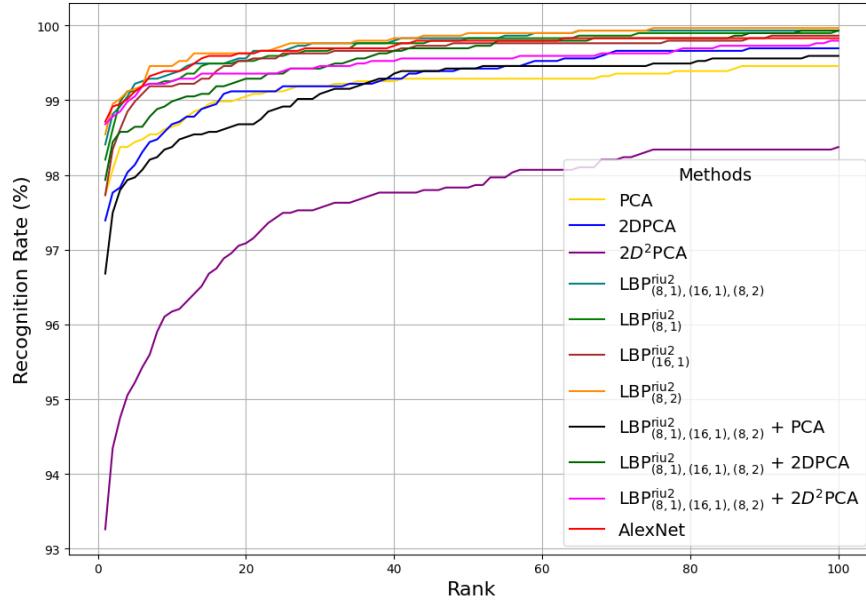


Figure 5.2: FVUSM Session Independent (S2_P2)

5.3 Comparative Analysis with Existing Methods

Table 5.5 presents a systematic comparison of the proposed methods with related studies on the FV-USM dataset under Strategy 2 across Protocols 1–3. Furthermore, the performance of the proposed approaches is compared with several CNN-based architectures on FV-USM, MMCBNU-6000, and UTFVP in Tables 5.6, 5.7, and 5.8. Overall, Table 5.9 summarizes all reported recognition accuracies, including representative prior work, CNN-based baselines, our study, and the proposed multi-radius fused $LBP^{\text{riu}2}_{(8,1),(16,1),(8,2)}$ variants across the three datasets and protocols. These comparisons demonstrate the competitiveness of the proposed multi-radius fused $LBP^{\text{riu}2}_{(8,1),(16,1),(8,2)}$ and its dimensionality-reduced variants against certain CNN-based methods.

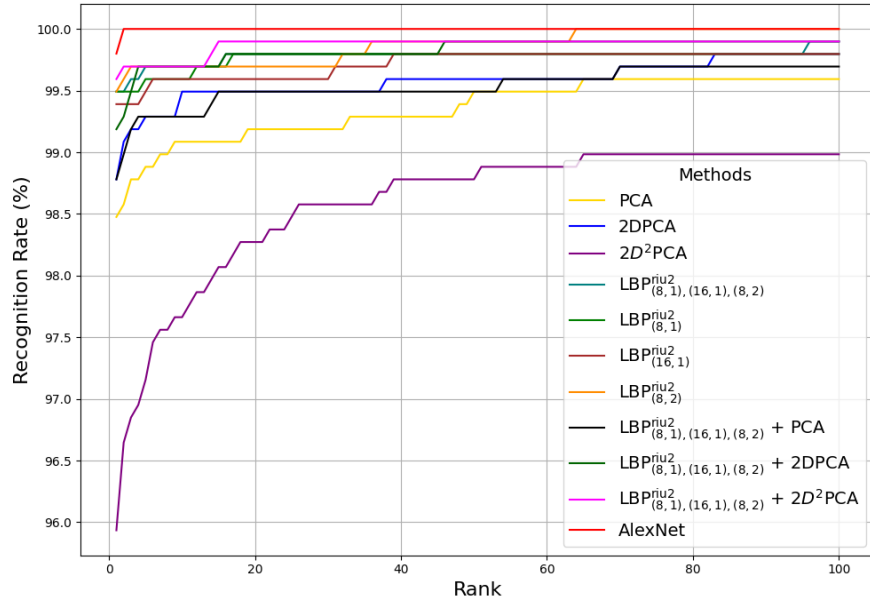


Figure 5.3: FVUSM Session Independent (S2_P3)

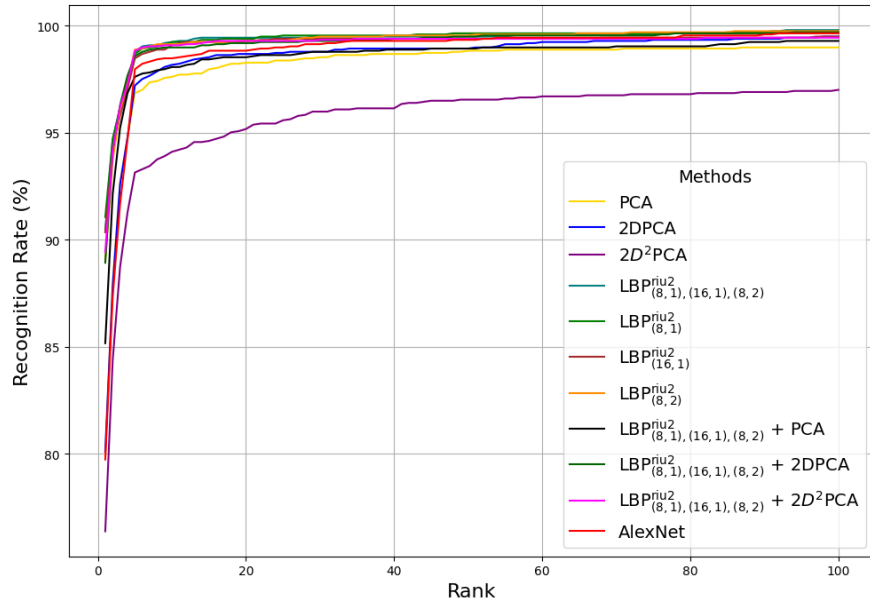


Figure 5.4: FVUSM Session Sensitive (S2_P1)

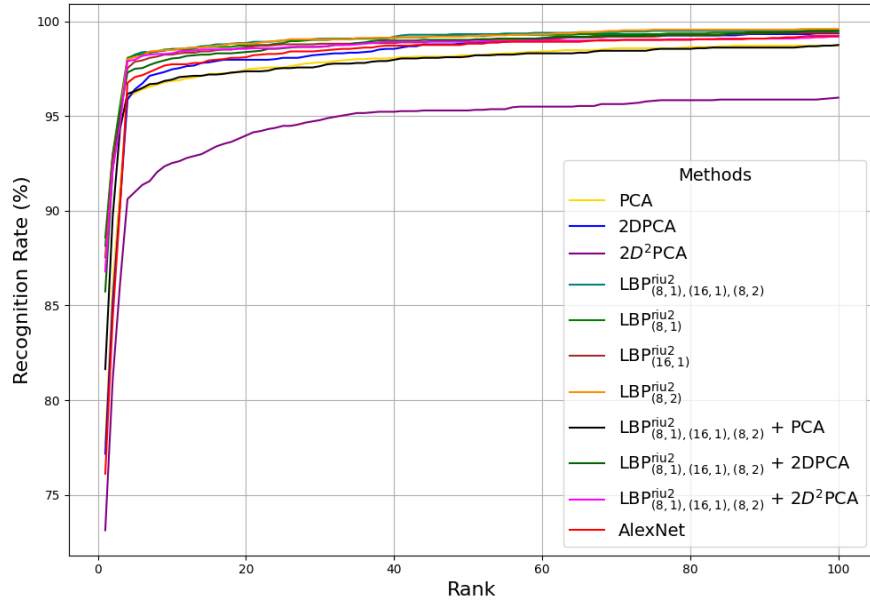


Figure 5.5: FVUSM Session Sensitive (S2_P2)

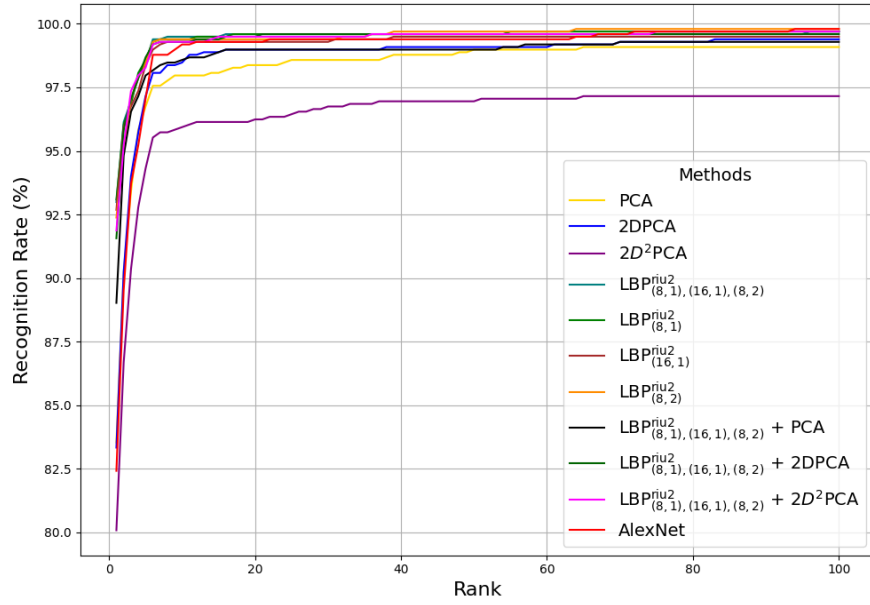


Figure 5.6: FVUSM Session Sensitive (S2_P3)

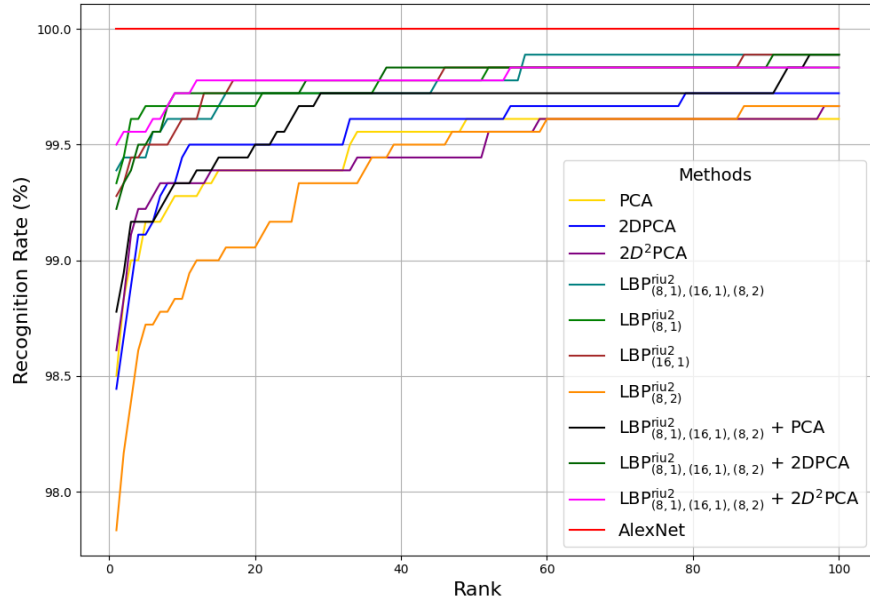


Figure 5.7: MMCBNU One Session Evaluation (S2_P1)

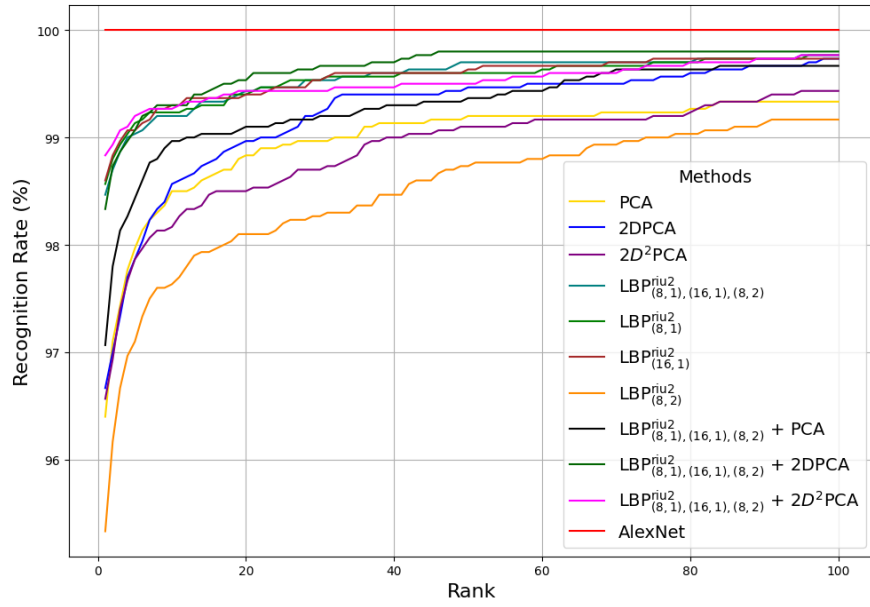


Figure 5.8: MMCBNU One Session Evaluation (S2_P2)

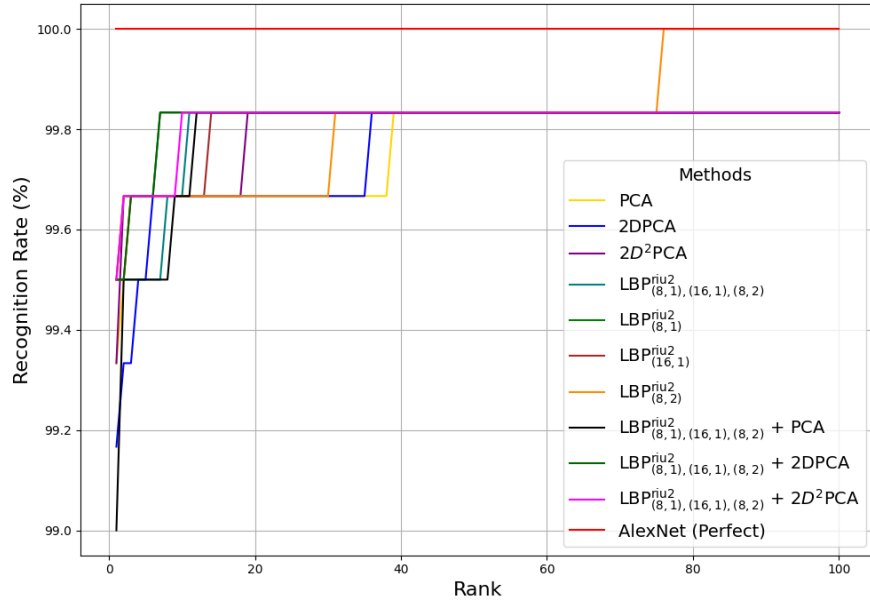


Figure 5.9: MMCBNU One Session Evaluation (S2_P3)

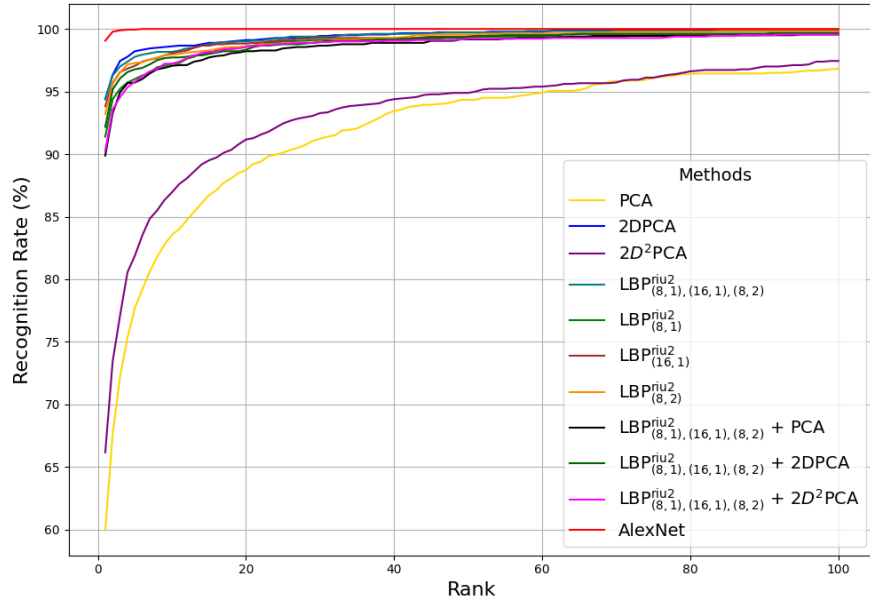


Figure 5.10: UTFVP One Session Evaluation (S2_P1)

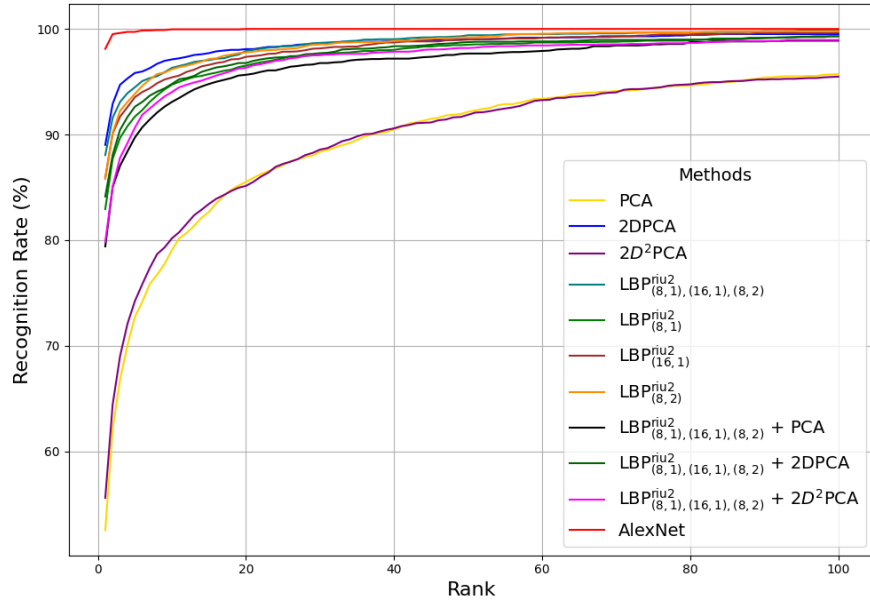


Figure 5.11: UTFVP One Session Evaluation (S2_P2)

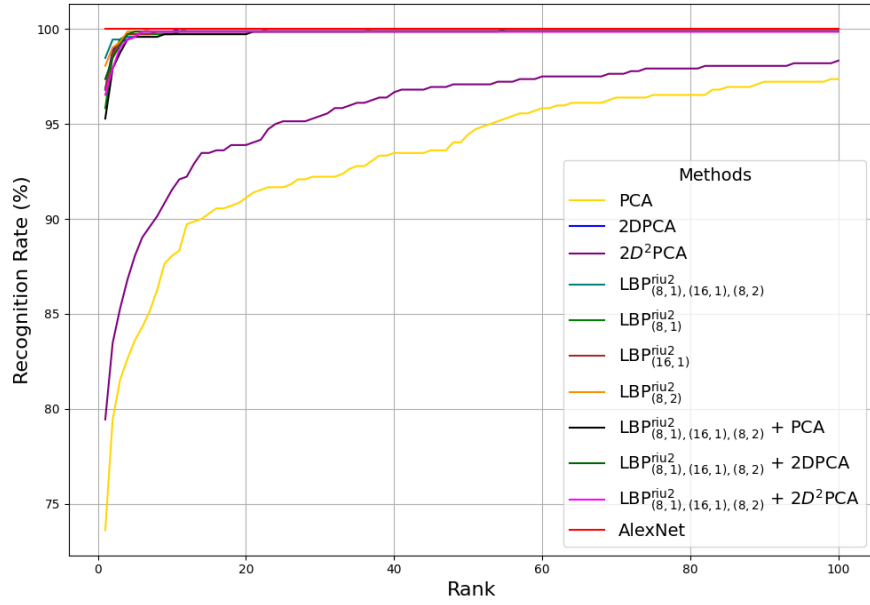


Figure 5.12: UTFVP One Session Evaluation (S2_P3)

Table 5.5: Comparison of Recognition Accuracy (%) under Strategy 2 Across Multiple Protocols on FV-USM Dataset

Sources	Methods	k = 3 (Protocol 2)	k = 4 (Protocol 1)	k = 5 (Protocol 3)
[19]	PCA	82.66	85.64	84.82
	2DPCA	82.83	94.99	97.56
	(2D) ² PCA	95.93	95.93	98.10
	MULBP + PCA	84.04	88.93	92.37
	MULBP + 2DPCA	98.01	98.64	99.46
	MULBP + (2D) ² PCA	97.83	98.19	98.57
[18]				
	LBP ^{u2} _(8,1)	75.34	76.42	77.78
	LBP ^{ri} _(8,1)	84.46	84.69	89.16
	LBP ^{riu2} _(8,1)	71.36	74.12	73.98
	LBP ^{u2} _(8,1) + PCA	–	–	85.45
	LBP ^{riu2} _(8,1) + PCA	–	–	85.37
	LBP ^{u2} _(8,1) + 2DPCA	–	–	95.75
	LBP ^{ri} _(8,1) + 2DPCA	–	–	96.03

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Table 5.5: Comparison of Recognition Accuracy (%) under Strategy 2 across Multiple Protocols on FV-USM Dataset (continued)

Source	Method	k = 3 (Protocol 2)	k = 4 (Protocol 1)	k = 5 (Protocol 3)
Our Study	$\text{LBP}_{(8,1)}^{\text{u2}} + 2\text{DPCA (w/o interp.)}$	–	–	98.10
	$\text{LBP}_{(8,1)}^{\text{ri}} + 2\text{DPCA (w/o interp.)}$	–	–	97.15
	PCA	97.73	97.76	98.48
	2DPCA	97.39	97.61	98.78
	$(2\text{D})^2\text{PCA}$	93.26	94.56	95.93
	$\text{LBP}_{(8,1)}^{\text{riu2}}$	98.73	98.20	99.49
	$\text{LBP}_{(16,1)}^{\text{riu2}}$	98.42	97.73	99.39
	$\text{LBP}_{(8,2)}^{\text{riu2}}$	98.93	98.54	99.49
	AlexNet	98.71	99.14	99.80
Proposed Methods	$\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}}$	98.41	98.73	99.49
	$\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}} + \text{PCA}$	96.68	97.76	98.78
	$\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}} + 2\text{DPCA}$	97.95	98.73	99.19
	$\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}} + (2\text{D})^2\text{PCA}$	98.68	99.19	99.59

Table 5.6: CNN-Based Comparisons on FV-USM

Methods	References	P1	P2	P3
CNN-based Methods	R. Das et al., 2018 [41]	98.58%	97.83%	–
	A. Dosovitskiy et al., 2020 [42]	95.99%	–	–
	D. Zhao et al., 2020 [43]	97.95%	–	–
	I. Boucherit et al., 2022 [44]	96.15%	94.21%	96.75%
	L. D. Tamang et al., 2022 [45]	97.84%	–	–
	N. C. Tran et al., 2023 [46]	–	96.77%	–
Proposed Methods	$\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}}$	98.73	98.41	99.49
	$\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}} + (2D)^2\text{PCA}$	98.68	99.19	99.59

Table 5.7: CNN-Based Comparisons on MMCBNU-6000

Methods	References	P1	P2	P3
CNN-Based Methods	K. He et al., 2016 [47]	99.50%	–	–
	F. Wang et al., 2017 [48]	99.17%	–	–
	A. Dosovitskiy et al., 2020 [42]	95.13%	–	–
Proposed Method	$\text{LBP}_{(8,1),(16,1),(8,2)}^{\text{riu2}} + 2\text{D}^2\text{PCA}$	99.50%	98.83%	99.50%

Table 5.8: CNN-Based Comparisons on UTFVP

Methods	References	P1	P2	P3
CNN-Based Methods	K. He et al., 2016 [47]	87.78%	–	–
	F. Wang et al., 2017 [48]	80.83%	–	–
	A. Dosovitskiy et al., 2020 [42]	63.47%	–	–
	W. Liu et al., 2023 [49]	93.47%	–	–
Proposed Method	LBP ^{riu2} _{(8,1),(16,1),(8,2)}	94.39%	88.06%	98.47%

Table 5.9: Unified Summary of Recognition Accuracies (%) Across FV-USM, MMCBNU-6000, and UTFVP Under Strategy 2 (Unreported Entries Are Denoted by “-”. For MMCBNU-6000 and UTFVP, only Protocol 1 Is Reported Here for Brevity.)

Source	Method	FV-USM			MMCBNU-6000			UTFVP		
		P1	P2	P3	P1	P2	P3	P1	P2	P3
[19]	PCA	85.64	82.66	84.82	-	-	-	-	-	-
	2DPCA	94.99	82.83	97.56	-	-	-	-	-	-
	(2D) ² PCA	95.93	95.93	98.10	-	-	-	-	-	-
	MULBP + PCA	88.93	84.04	92.37	-	-	-	-	-	-
	MULBP + 2DPCA	98.64	98.01	99.46	-	-	-	-	-	-
	MULBP + (2D) ² PCA	98.19	97.83	98.57	-	-	-	-	-	-
[18]	$LBP_{(8,1)}^{u2}$	76.42	75.34	77.78	-	-	-	-	-	-
	$LBP_{(8,1)}^r$	84.69	84.46	89.16	-	-	-	-	-	-
	$LBP_{(8,1)}^{riu2}$	74.12	71.36	73.98	-	-	-	-	-	-
	$LBP_{(8,1)}^{u2} + PCA$	-	-	85.45	-	-	-	-	-	-
	$LBP_{(8,1)}^{riu2} + PCA$	-	-	85.37	-	-	-	-	-	-
	$LBP_{(8,1)}^{u2} + 2DPCA$	-	-	95.75	-	-	-	-	-	-
	$LBP_{(8,1)}^r + 2DPCA$	-	-	96.03	-	-	-	-	-	-
		-	-	-	-	-	-	-	-	-

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Table 5.9: Unified summary of recognition accuracies (%) across FV-USM, MMCBNU-6000, and UTFVP under Strategy 2 (continued).

Source	Method	FV-USM			MMCBNU-6000			UTFVP
		P1	P2	P3	P1	P1	P1	
	$\text{LBP}_{(8,1)}^{u2} + 2\text{DPCA (w/o interp.)}$	–	–	98.10	–	–	–	–
	$\text{LBP}_{(8,1)}^{ri} + 2\text{DPCA (w/o interp.)}$	–	–	97.15	–	–	–	–
[41]	CNN-based method	98.58	97.83	–	–	–	–	–
[42]	ViT	95.99	–	–	95.13	–	63.47	–
[43]	LCNN-CL-DR	97.95	–	–	–	–	–	–
[44]	Merge-CNN	96.15	94.21	96.75	–	–	–	–
[45]	Hybrid Pooling CNN	97.84	–	–	–	–	–	–
[46]	CNN-based method	–	96.77	–	–	–	–	–
[47]	ResNet50	–	–	–	99.50	–	87.78	–
[48]	RAN	–	–	–	99.17	–	80.83	–
[49]	MMRAN	–	–	–	–	–	93.47	–

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Table 5.9: Unified summary of recognition accuracies (%) across FV-USM, MMCBNU-6000, and UTFVP under Strategy 2 (continued).

Source	Method	FV-USM			MMCBNU-6000			UTFVP		
		P1	P2	P3	P1	P2	P3	P1	P2	P3
Our study	PCA	97.76	97.73	98.48	98.50			59.94		
	2DPCA	97.61	97.39	98.78	98.44			94.39		
	(2D) ² PCA	94.56	93.26	95.93	98.61			66.17		
	$\text{LBP}_{(8,1)}^{\text{riu}2}$	98.73	98.20	99.49	99.33			91.39		
	$\text{LBP}_{(16,1)}^{\text{riu}2}$	98.42	97.73	99.39	99.28			93.83		
	$\text{LBP}_{(8,2)}^{\text{riu}2}$	98.93	98.54	99.49	97.83			93.22		
	AlexNet	99.14	98.71	99.80	100.00			99.06		

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Table 5.9: Unified summary of recognition accuracies (%) across FV-USM, MMCBNU-6000, and UTFVP under Strategy 2 (continued).

Source	Method	FV-USM			MMCBNU-6000			UTFVP		
		P1	P2	P3	P1	P2	P3	P1	P2	P3
Proposed methods	$\text{LBP}^{\text{riu2}}_{(8,1),(16,1),(8,2)}$	98.73	98.41	99.49	99.39			94.39		
	$\text{LBP}^{\text{riu2}}_{(8,1),(16,1),(8,2)} + \text{PCA}$	97.76	96.68	98.78	98.78			89.89		
	$\text{LBP}^{\text{riu2}}_{(8,1),(16,1),(8,2)} + 2\text{DPCA}$	98.73	97.95	99.19	99.22			92.17		
	$\text{LBP}^{\text{riu2}}_{(8,1),(16,1),(8,2)} + (2\text{D})^2\text{PCA}$	99.19	98.68	99.59	99.50			90.28		

Chapter 6

CONCLUSION AND FUTURE WORKS

This thesis demonstrates that finger-vein identification is a practical and effective biometric technique. Using near-infrared imaging, it captures the unique patterns of veins beneath the skin, which makes it reliable for distinguishing individuals and relatively difficult to spoof. To study this area in a structured way, a complete evaluation framework was designed and tested, covering a broad range of methods—from classical dimensionality-reduction pipelines combined with texture descriptors to a modern deep learning model.

Experiments were conducted on three public datasets (FV-USM, MMCBNU-6000, and UTFVP) using three standard protocols and two fusion strategies. This setup makes it possible to clearly observe how each method performs under real-world challenges such as session differences, limited training data, and varying image quality. In this work, the multi-radius-fused rotational-invariant uniform Local Binary Pattern descriptor ($\text{LBP}^{\text{riu}^2}_{(8,1)(16,1)(8,2)}$) paired with dimensionality-reduction techniques such as Principal Component Analysis (PCA), Two-Dimensional PCA (2DPCA), and Two-Directional Two-Dimensional ($2\text{D}^2\text{PCA}$), produced feature sets that were both compact and easy to interpret. These hand-crafted representations stayed highly competitive, particularly in the mid and high rank regions of the Cumulative Match Characteristic (CMC) curves, and showed clear strengths in low-data settings or under session-sensitive conditions. This pipeline showed competitive performance when compared with current state-of-the-art methods and, in several cases, slightly outperformed them.

On the contrary, when sufficient training data were available, the customized AlexNet

model consistently achieved higher rank-1 identification accuracy. By learning complex hierarchical patterns directly from the images, the network performed exceptionally well in session-independent setups and in protocols with larger enrollment sets, approaching near-perfect recognition.

Strategy 1 and Strategy 2 also showed clear differences. With Strategy 1, information from multiple fingers is combined at the identity level, which usually leads to stronger subject-level recognition. With Strategy 2, each finger is matched on its own, so finger-specific details are kept instead of being mixed with other samples. This is useful when the application needs a per-finger decision or when computation is limited.

The CMC curves also show a consistent trend: the deep model performs best at the earliest ranks, while hand-crafted descriptors improve more gradually as the gallery size grows. Finally, the results make it clear that performance depends not only on the algorithm, but also on how the protocols are defined and on the dataset itself. Differences between sessions, changes in image quality, and demographic variation can all affect accuracy. These factors should therefore be considered when selecting a recognition pipeline, since they influence when hand-crafted methods are more suitable and when deep learning becomes the better option.

Future work can build on this thesis in several practical ways. First, evaluation should place more weight on verification-style testing (one-to-one matching) alongside identification, using metrics such as EER and DET curves to better reflect real biometric operation. Another priority is improving robustness to cross-session variation, since changes in placement, illumination, and capture conditions can shift performance, especially when training data are limited; stronger normalization, alignment, and session-aware or domain-adaptation training could help. Hybrid designs that combine hand-crafted descriptors with deep features are also worth deeper study, because they may balance the stability of classical cues with the

representational power of deep models, particularly in small-data settings.

In addition, metaheuristic optimization methods (such as Genetic Algorithms or PSO) could take much of the guesswork out of system design by automatically picking the most useful feature components, helping decide how fusion should be done, and searching for better model settings or network designs. Finally, future work should pay more attention to template security by exploring cancelable biometrics, secure feature transformations, and biometric key-binding, so that stored templates are harder to misuse while the system still keeps good recognition accuracy.

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